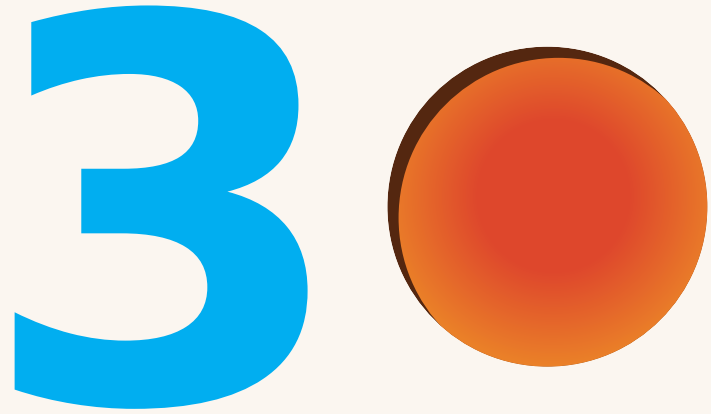


STRESS CONTROL

6 STRESS CONTROL PROTOCOLS

1. Do One to Three Physiological Sighs to Reduce Stress Fast
2. Use Deliberate Cold Exposure to Raise Your Stress Threshold
3. Do Non-Sleep Deep Rest (NSDR) to Buffer Against Stress
4. Do a Micro Meditation Immediately After Waking Up
5. Boost Your Morning Cortisol and Catecholamine Peak
6. Consider Specific Supplements to Help Offset Excessive Stress

I, like everyone else, don’t like the feeling of being stressed, but I have grown accustomed to the idea that life inevitably involves stress. Whether it was my high school classmates (most of whom were very driven), the athletes I’ve met, or the students, professors, and people in tech or creative fields around me, I have always found that stress accompanies ambition, and that for many of the top achievers, it’s a constant.

The weird thing is that many of the unambitious people I have known often said they were stressed out too. Lack of direction, lack of money, the feeling they were slipping behind in life, health issues, and more… Eventually I had to conclude that this was indeed part of life—like breathing, sleeping, or eating. That meant there had to be a good way to address it.

The problem was that when I looked to see what the science said about stress, it was uniformly negative. It made clear that stress wasn’t just uncomfortable, it was damaging to the brain and body. And that only made stress more stressful. Moreover, the whole framing of stress seemed wrong, or at least rather one-sided.

For instance, “fight or flight” is one of the most common phrases we hear when the topic of stress comes up. It’s easy to remember and it seems science-y—because it sounds like it could describe how the brain and body respond to stress.

Unfortunately, it's more misleading than informative. We also often hear that the stress response is an inconvenient holdover from our evolutionary past. Meaning, the uncomfortable state of arousal we call stress was helpful for us long ago, propelling us to escape from saber-toothed tigers and other large predators, but it's no longer useful.

I always appreciate when science shows up in a conversation, but I confess phrases like "fight or flight" make me shake my head. It's not just that they are misleading; they also don't provide any valuable information about how to navigate the stress response. And actually, the stress response is no holdover. And research has shown that the beliefs we hold about stress can change how stress impacts our health, for better or for worse, so we need to be careful what we tell ourselves about it.[1]

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[1] Modupe Akinola et al., "Adaptive Appraisals of Anxiety Moderate the Association Between Cortisol Reactivity and Performance in Salary Negotiations," PLOS One 11, no. 12 (December 2016): e0167977, https://doi.org/10.1371/journal.pone.0167977; Alia J. Crum et al., "Rethinking Stress: The Role of Mindsets in Determining the Stress Response," Journal of Personality and Social Psychology 104, no. 4 (April 2013): 716–33, https://doi.org/10.1037/a0031201; Alia Crum et al., "The Role of Stress Mindset in Shaping Cognitive, Emotional, and Physiological Responses to Challenging and Threatening Stress," Anxiety, Stress, & Coping 30, no. 4 (January 2017): 379–95, https://doi.org/10.1080/10615806.2016.1275585; Alia J. Crum et al., "Optimizing Stress: An Integrated Intervention for Regulating Stress Responses," Emotion 20, no. 1 (February 2020): 120–25, https://doi.org/10.1037/emo0000670; Alia J. Crum et al., "Evaluation of the 'Rethink Stress' Mindset Intervention: A Metacognitive Approach to Changing Mindsets," Journal of Experimental Psychology 153, no. 9 (2023): 2603–22, http://doi.org/10.1037/xge0001396; Roman Dunce et al., "Acute Exposure to Stress Improves Performance in Trace Eyeblink Conditioning and Spatial

Stress is an area I've thought a lot about from a neuroscience perspective. My lab has studied it in animals and humans. Decades into my career, I now know that there is, in fact, a good way to address it. There are excellent stress management tools based in science. So, given how pervasive stress is, I am especially excited to share the information in this chapter.

First let's begin with the facts. Forget saber-toothed tigers. Many of the same situations that stress us out now have been stressing out humans for thousands of years. An argument with a loved one, a persistent pain of unknown cause, a worry about a child, a looming payment, and on and on. Scenarios like these are not unique to modern times.

A good way to think about stress is this: Stress is nature's attempt to create a single solution to all matters that might be, or actually are, urgent. I'll say that again. Stress is nature's attempt to create a single solution to all matters that might be, or actually are, urgent.

Learning Tasks in Healthy Men," Learning & Memory 14, no. 5 (May 2007): 329–35, https://doi.org/10.1101/lm.483807; Jeremy P. Jamieson et al., "Mind over Matter: Reappraising Arousal Improves Cardiovascular and Cognitive Responses to Stress," Journal of Experimental Psychology: General 141, no. 3 (September 2011): 417–22, https://doi.org/10.1037/a0025719; Eric N. Smith et al., "Stress, Mindsets, and Success in Navy SEALs Special Warfare Training," Frontiers in Psychology 10 (January 2020): 2962, https://doi.org/10.3389/fpsyg.2019.02962; Allison Wood Brooks, "Get Excited: Reappraising Pre-Performance Anxiety as Excitement," Journal of Experimental Psychology: General 143, no. 3 (2014): 1144–58, https://doi.org/10.1037/a0035325.

So no matter what the stressor (and there are an infinite number of them out there), the basic architecture of the stress response—that is, what happens to our brain and body when we perceive a stressor—is the same. Cortisol and other hormones, including the catecholamine neuromodulators adrenaline and norepinephrine, flood our system. Adrenaline, or epinephrine, as well as cortisol are released from our adrenal glands. Norepinephrine is released from specific locations in the brain including the locus coeruleus, which we discussed in chapter 1 and chapter 2. Dopamine is released in the brain and in the body too. All of these chemicals change the activity level of specific neuronal circuits and, in doing so, initiate a coordinated shift in the ways our organs function, and thus in our overall state of mind and body. We often hear about the increase in heart rate, breathing, and blood pressure that stress brings about. Less discussed, however, are the perceptual changes. Stress changes our visual field and narrows our mental attention.[2] We become fixated on the events related to the stressor: "How did this happen? How can I undo this?!" Stress causes our perception of time to slow down. Seconds start to feel like minutes, minutes like hours, hours like days… Meanwhile, our rate of thinking—the sheer number of thoughts we have per second—seems to speed up. And yet,

[2] J. A. Easterbrook, "The Effect of Emotion on Cue Utilization and the Organization of Behavior," Psychological Review 66, no. 3 (May 1959): 183–201, https://doi.org/10.1037/h0047707; Robert J. Hockey, "Stress and the

our cognitive flexibility is reduced. We default to simple black-and-white thinking. Nuance evaporates. Mild-to-moderate stress makes us feel jittery whereas intense stress can cause us to freeze. At its extreme, it can make people feel as if they have left their body, as if they are viewing themselves from above.[3]

While that all sounds dreadful, we must remember that the stress response is adaptive. The purpose of it is to put our brain and body into a state that gives us the best chance of addressing the problem at hand. A single solution. In an ideal world, the degree of stress response would perfectly match the stressor we face. Running late? A little bit more arousal to help us focus on the key items we need to get out

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Cognitive Components of Skilled Performance," in Human Stress and Cognition: An Information Processing Approach, ed. Vernon Hamilton (John Wiley & Sons, 1979); Lorraine Hope et al., "Witnesses in Action: The Effect of Physical Exertion on Recall and Recognition," Psychological Science 23, no. 4 (2012): 386–90, https://doi.org/10.1177/0956797611431463; Jessica Wapner, "Secrets to Surviving Stressful Times," Scientific American Mind 32, no. 1 (January 2021): 14, https://www.scientificamerican.com/article/vision-and-breathing-may-be-the-secrets-to-surviving-2020/; R. J. Miller and Masatoshi Takahama, "Effects of Relaxation and Aversive Visual Stimulation on Dark Focus Accommodation," Ophthalmic and Physiological Optics 7 (July 1987): 219–23, https://doi.org/10.1111/j.1475-1313.1987.tb00736.x; Edward C. Godnig, "Tunnel Vision: Its Causes and Treatment Strategies," Journal of Behavioral Optometry 14, no. 4 (2003): 95–99, https://www.oepf.org/wp-content/uploads/2021/08/14-420Godnig1.pdf.

[3] Lindsay D. Salay et al., "A Midline Thalamic Circuit Determines Reactions to Visual Threat," Nature 557, no. 7704 (May 2018): 183–89, https://doi.org/10.1038/s41586-018-0078-2; Joseph E. LeDoux and Daniel S. Pine, "Using Neuroscience to Help Understand Fear and Anxiety: A Two-System Framework," American Journal of Psychiatry 173, no. 11 (November 2016): 1083–93, https://doi.org/10.1176/appi.ajp.2016.16030353.

the door with grace would be perfect. See our toddler running toward the traffic? A huge surge in energy to quickly grab them is what we want! In fact, the parts of our nervous system that govern stress—called the autonomic nervous system—can be calibrated to generally match our stress level to the stressor we face, from mild to severe, from brief to ongoing.[4] But nowadays, the many comforts most people enjoy thanks to technology—indoor heating and air conditioning, abundant food and shelter, and all in all our general lack of actual survival stressors—have our autonomic nervous systems uncalibrated. We are far more labile under the constant barrage of minor psychological stressors all day and night, and a lot of people are in a constant state of stress related to things they simply cannot control or influence even slightly. Put simply, the frequency and/or magnitude of the stress response is far too high relative to the stressors many people face. The occasional bout of stress lasting a few hours or less is no big deal, and even a few days is something our bodies can handle without

[4] Huda Akil et al., “Stress Fitness: A Neuroscientific Approach to Building Emotional Resilience,” Neuron (2026), https://doi.org/10.1016/j.neuron.2026.03.032; Bruce S. McEwen, “Stress, Adaptation, and Disease: Allostasis and Allostatic Load,” Annals of the New York Academy of Sciences 840 (1998): 33–34, https://doi.org/10.1111/j.1749-6632.1998.tb09546.x; Rachel Gilgoff et al., “The Stress Phenotyping Framework: A Multidisciplinary Biobehavioral Approach for Assessing and Therapeutically Targeting Maladapted Stress Physiology,” Stress 27, no. 1 (January 2024): 2327333, https://doi.org/10.1080/10253890.2024.2327333.

long-term consequences. Once we experience a few weeks to a month or more of daily (which usually also means nightly) stress, that's when we start to suffer. Chronically elevated resting blood pressure and heart rate, not sleeping well or enough due to rumination or feeling "tired but wired," and other shifts to our internal state become detrimental to our immediate and long-term health. We've all heard the laundry list of things that stress harms: our hormones, memory, mood, and decision-making ability, as well as the many diseases it makes worse or can bring about to our heart, brain, gut, and liver. The body of evidence that chronic stress is bad for us is vast and robust.

Yet some stress is good for us. In fact, the occasional spike in our stress hormones can increase testosterone[5] and improve our immune system functioning during the hours that follow.[6] In the short term, it can improve our memory.[7] The problems arise when the intensity or the duration of the stress response is too great.

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[5] Regula Zueger et al., "Testosterone and Cortisol Responses to Acute and Prolonged Stress During Officer Training School," Stress 26, no. 1 (2023), https://doi.org/10.1080/10253890.2023.2199886; Yaxin Fan et al., "Salivary Testosterone and Cortisol Response in Acute Stress Modulated by Seven Sessions of Mindfulness Meditation in Young Males," Stress 27, no. 1 (2024), https://doi.org/10.1080/10253890.2024.2316041.

[6] Amna Alotiby, "Immunology of Stress: A Review Article," Journal of Clinical Medicine 13, no. 21 (2024): 6394, https://doi.org/10.3390/jcm13216394.

[7] Brynn E. Sherman et al., "Hippocampal Mechanisms Support Cortisol-Induced Memory Enhancements," Journal of Neuroscience 43, no. 43 (October 2023): JN-23, https://doi.org/10.1523/jneurosci.0916-23.2023; William J. Kraemer and Nicholas A. Ratamess, "Hormonal Responses and

Eliminating stress from our lives is unrealistic. So our goal with stress should be twofold. First, we need to be able to adjust the intensity and the duration of our stress response so we can better handle stress in the moment. Second, many of us need to increase our stress threshold—that is, the subjective point at which events and thoughts trigger large amounts of adrenaline release in our body—so our stress response is calibrated to any stressors we may encounter. Ahem, I mean, that we will encounter. When adrenaline release occurs as the consequence of tiny things, that is not good. It negatively affects our physical health and

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Adaptations to Resistance Exercise and Training," Sports Medicine 35, no. 4 (2005): 339–61, https://doi.org/10.2165/00007256-200535040-00004; Truls Raastad et al., "Hormonal Responses to High- and Moderate-Intensity Strength Exercise," European Journal of Applied Physiology 82 (May 2000): 121–28, https://doi.org/10.1007/s004210050661; David C. Cumming et al., "Reproductive Hormone Responses to Resistance Exercise," Medicine & Science in Sports & Exercise 19, no. 3 (June 1987): 234–38; Stuart F. White et al., "Putting the Flight in 'Fight-or-Flight': Testosterone Reactivity to Skydiving Is Modulated by Autonomic Activation," Biological Psychology 143 (April 2019): 93–102, https://doi.org/10.1016/j.biopsycho.2019.02.012; S. D. Richter et al., "Time Kinetics of the Endocrine Response to Acute Psychological Stress," Journal of Clinical Endocrinology and Metabolism 81, no. 5 (May 1996): 1956–60, https://doi.org/10.1210/jcem.81.5.8626864; F. S. Dhabhar and B. S. McEwen, "Acute Stress Enhances While Chronic Stress Suppresses Cell-Mediated Immunity In Vivo: A Potential Role for Leukocyte Trafficking," Brain, Behavior, and Immunity 11, no. 4 (December 1997): 286–306, https://doi.org/10.1006/brbi.1997.0508; Firdaus S. Dhabhar, "Enhancing Versus Suppressive Effects of Stress on Immune Function: Implications for Immunoprotection and Immunopathology," Neuroimmunomodulation 16, no. 5 (2009): 300–317, https://doi.org/10.1159/000216188; Firdaus S. Dhabhar, "Effects of Stress on Immune Function: The Good, the Bad, and the Beautiful," Immunologic Research 58 (2014): 193–210, https://doi.org/10.1007/s12026-014-8517-0; Suzanne C. Segerstrom and Gregory E. Miller, "Psychological Stress and the Human Immune System: A

relationships, including our relationship with ourselves. Do you like being around you when you're stressed? Didn't think so. We can change our relationship to stress through cognitive reframing, but an effective approach is to learn to quickly control our stress response once it hits, and to periodically reset our stress threshold to a level that best serves us. We can't always control what life throws at us, but we can indeed control our stress response, and we can raise our stress threshold. That's what this chapter is all about.

I first became interested in investigating the neuroscientific mechanisms behind stress back in 2015. At the time, grit, resilience, and mental toughness were popular themes in books and media. And those are great qualities. The examples—both historical and contemporary—of people being gritty and resilient and mentally tough are there to

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Meta-Analytic Study of 30 Years of Inquiry," Psychological Bulletin 130, no. 4 (2004): 601–30, https://doi.org/10.1037/0033-2909.130.4.601; Larry Cahill et al., "β-Adrenergic Activation and Memory for Emotional Events," Nature 371 (1994): 702–4, https://doi.org/10.1038/371702a0; Marian Joëls et al., "Learning Under Stress: How Does It Work?," Trends in Cognitive Sciences 10, no. 4 (April 2006): 152–58, https://doi.org/10.1016/j.tics.2006.02.002; Carmen Sandi and M. Teresa Pinelo-Nava, "Stress and Memory: Behavioral Effects and Neurobiological Mechanisms," Neural Plasticity 2007 (2007): 78970, https://doi.org/10.1155/2007/789; Tom Smeets et al., "True or False? Memory Is Differentially Affected by Stress-Induced Cortisol Elevations and Sympathetic Activity at Consolidation and Retrieval," Psychoneuroendocrinology 33, no. 10 (November 2008): 1378–86; Susanne Vogel and Lars Schwabe, "True or False? Memory Is Differentially Affected by Stress-Induced Cortisol Elevations and Sympathetic Activity at Consolidation and Retrieval," Psychoneuroendocrinology 71 (September 2016): 64–72, https://doi.org/10.1016/j.psyneuen.2016.04.027.

remind us that humans harbor within them an enormous capacity to manage and even channel their stress.

While there is a lot known about the neural circuits and hormones that control the stress response, I specifically wanted to understand the mechanisms that determine its intensity and duration. I wanted to know which features of the stress response, like heart rate, blood pressure, attentional narrowing, and so on, were amenable to adjustment in real time. I also wanted to know whether there are levers for adjusting the duration, the intensity, and maybe even the probability of the stress response getting triggered in the first place.

I was able to find some data about stress inoculation,[8] including some studies supporting the idea that certain forms of early-life stress can set the stage for resilience later in life by virtue of how they tune the physiology of the stress response to better match the intensity of a stressor.[9] But it was hard to find evidence for specific tools that kids or adults could reliably apply to recalibrate the stress response

[8] Donald H. Meichenbaum and Jerry L. Deffenbacher, “Stress Inoculation Training,” Counseling Psychologist 16, no. 1 (1988): 69–90, https://doi.org/10.1177/0011000088161005; Zhanying Sun et al., “Preventing and Mitigating Post-Traumatic Stress: A Scoping Review of Resilience Interventions for Military Personnel in Pre-Deployment,” Psychology Research and Behavior Management 17 (June 17, 2024): 2377–89, https://doi.org/10.2147/PRBM.S459220.

[9] David M. Lyons et al., “Preliminary Evidence That Hippocampal Volumes in Monkeys Predict Stress Levels of Adrenocorticotropic Hormone,” Biological Psychiatry 62 (2007): 1171–74,

across different circumstances. I wanted to find mechanistic data that could be translated broadly and that were accessible for the average person.

I continued my search. I found studies on the benefits of positive self-talk[10] but just as many studies on its lack of effectiveness.[11] Despite the fact that we are all born with the same biological stuff—adrenals that release adrenaline, locus coeruleus neurons that release norepinephrine, and prefrontal cortices that can exert top-down control over our impulses and feeling states—I just couldn't find any really convincing studies in humans that pointed to tools people could use to improve their capacity to manage stress. Meanwhile, we know that slogans like "Just do it" and

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https://doi.org/10.1016/j.biopsych.2007.03.012; Steven F. Maier and Martin E. P. Seligman, "Learned Helplessness at Fifty: Insights from Neuroscience," Psychological Review 123, no. 4 (2016): 349–67, https://doi.org/10.1037/rev0000033; Teri Saunders et al., "The Effect of Stress Inoculation Training on Anxiety and Performance," Journal of Occupational Health Psychology 1, no. 2 (1996): 170–86, https://doi.org/10.1037/1076-8998.1.2.170; Glenn E. Richardson, "The Metatheory of Resilience and Resiliency," Journal of Clinical Psychology 58, no. 3 (March 2002): 307–21, https://doi.org/10.1002/jclp.10020.

[10] E. Kross et al., "Self-Talk as a Regulatory Mechanism: How You Do It Matters," Journal of Personality and Social Psychology 106, no. 2 (2014): 304–24, https://doi.org/10.1037/a0035173; Claire Eagleson et al., "The Power of Positive Thinking: Pathological Worry Is Reduced by Thought Replacement in Generalized Anxiety Disorder," Behaviour Research and Therapy 78 (March 2016): 13–18, https://doi.org/10.1016/j.brat.2015.12.017.

[11] Joanne V. Wood et al., "Positive Self-Statements: Power for Some, Peril for Others," Psychological Science 20, no. 7 (2009): 860–66, https://doi.org/10.1111/j.1467-9280.2009.02370.x; Eagleson et al., "Power of Positive Thinking."

mantras like “Embrace the suck,” or telling ourselves “You got this,” can be good for pushing us through a stressor we have chosen to face—a long run, the final reps of a hard set in the gym, a study session or big exam—but most people would agree that self-talk fails the vast majority of the time when the stressor isn’t one we’ve chosen, or when our efforts have us falling short of our goals. And at some point, real life brings the sorts of stressors we simply cannot control, like a medical emergency or the loss of a beloved friend or family member. In those circumstances, “Just do it” just doesn’t do it.

Of course, much has been researched and written about what I call offline practices to manage stress. Meditation, massages, vacations, hot baths, deep breaths, talk therapy. Great! Who doesn’t love a massage at the end of a hard day? Meditation is powerful as a tool to buffer against the stress response; a randomized clinical trial conducted by researchers at Georgetown University found that people with generalized anxiety disorder who meditated experienced a noteworthy reduction in pro-inflammatory cytokines—chemical messengers essential to our immune system but that can also trigger a stress response—compared to those who didn’t meditate.[12] Other studies, including one

[12] Elizabeth A. Hoge et al., “The Effect of Mindfulness Meditation Training on Biological Acute Stress Responses in Generalized Anxiety Disorder,” Psychiatry Research 262 (April 2018): 328–32, https://doi.org/10.1016/j.psychres.2017.01.006.

done by my lab at Stanford in collaboration with my colleague David Spiegel's lab, also at Stanford, have shown clear benefits of meditation practices as short as five minutes per day on stress, mood, and sleep.[13] That might seem like a trivial level of time commitment, especially given the many benefits of meditation.[14]

But I wanted to know how people can navigate stress while they are in the throes of the stress response. When their pulse is racing, their vision is blurry, their hands are shaking, and all they can think about is the stressor and the potential bad outcomes from it. That's when the capacity to manage stress matters most. Not offline, but in the moment.

As I pointed out above, the stress response is universal. Whatever the specific stressor might be, the stress response is the same. It varies in degree, not kind. This suggests that if we have reliable tools to reduce the stress response in one context, we ought to be able to reduce it in another. The

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[13] Julia C. Basso et al., "Brief, Daily Meditation Enhances Attention, Memory, Mood, and Emotional Regulation in Non-Experienced Meditators," Behavioural Brain Research 356 (January 1, 2019): 208–20, https://doi.org/10.1016/j.bbr.2018.08.023; Simon B. Goldberg et al., "Testing the Efficacy of a Multicomponent, Self-Guided, Smartphone-Based Meditation App: Three-Armed Randomized Controlled Trial," JMIR Mental Health 7, no. 11 (November 2020): e23825, https://doi.org/10.2196/23825; Melis Yilmaz Balban et al., "Brief Structured Respiration Practices Enhance Mood and Reduce Physiological Arousal," Cell Reports Medicine 4, no. 1 (January 2023): 1–15, https://doi.org/10.1016/j.xcrm.2022.100895.

[14] Daniel Goleman and Richard J. Davidson, Altered Traits: Science Reveals How Meditation Changes Your Mind, Brain and Body (Avery, 2017).

problem? No one had dissected the universal features of stress control, only the stress response.

One valuable thing everyone should know about the stress response, though, is that it is asymmetric: It starts extremely fast and turns off relatively slowly. The adrenaline release that is the first domino of the stress response happens in seconds or less and can continue for several minutes after we perceive a stressor.[15] Think of a time when you were groggy from just waking up and you picked up your phone to see a disturbing text. BOOM! You are wide awake and ready to move in an instant. That's the speed of adrenaline in action.

Then, even after the stressor is resolved, the stress response persists. Imagine getting a text that says, "I think Mom was on the flight that just crashed," and ten minutes later, "I just spoke to her, she wasn't on there. She's fine." Your feeling of relief will no doubt outpace how quickly your body settles down. That's an extreme example, but the same thing applies when someone almost sideswipes you while driving, and you swerve just in time to avoid a collision. A shot of adrenaline wicks out through your body in an instant, and almost as fast, norepinephrine is released to ramp up attentional circuits in your brain. Cognitively, you realize you're safe almost as quickly as you thought you were about

[15] Hannes Ernst et al., "Assessment of the Human Response to Acute Mental Stress—an Overview and a Multimodal Study," PLOS One 18, no. 11 (2023): e0294069, https://doi.org/10.1371/journal.pone.0294069.

to get hit, but your body and mind don't relax immediately. More often than not, we plummet into the stress response and it takes time to return to baseline. And that's when everything turns out okay. When stressors persist, you experience wave after wave of adrenaline. And while the basic circuits for stress are the same in everyone, seldom discussed is that not everyone has the same threshold, or trigger point, for a stress response to kick off. For some people it's as if that first domino is fixed in cement. Such people are rare. For most people, however, the domino falls too easily. I wanted to understand this aspect of stress too, in order to develop tools anyone can use to raise their stress threshold, to allow them to calibrate their autonomic set point. That is, the balance point that our nervous system works to maintain—so that the stress response happens when it's needed and not when it isn't. Even better would be to find ways such that when stress does happen, it occurs in proportion to the stressor. I wanted these tools to be mechanistic and behavioral, not pharmacological, because pharmacological tools—while they can be helpful—don't allow for real-time adjustments. You swallow a pill and that's that. And they have to be taken before stress arrives, which is problematic because we don't always know when stressors are coming. I also wanted to develop tools that do not rely on practice, like cognitive behavioral therapy, which

is great but can take a fair amount of time to master[16] and doesn't always translate across stressors very well; there is evidence both for and against generalizability.[17] Like all protocols in this book, I wanted tools for stress management that could be effective the first time and every time because they are anchored in hardwired circuits and processes we all possess. So I dove into the research on the physiology of stress to come up with a different way to frame the whole concept of stress. And stress control. That led to…

The Seesaw Model

Think about the stress response as existing somewhere along a spectrum of arousal, with very low arousal at one end of the spectrum and very high arousal at the other. At the extremely low arousal end is a comatose state. Slightly more aroused than that would be deep sleep, then light sleep. Then toward the middle of the spectrum is a state in which we are alert but calm (which incidentally is the ideal state for most

[16] Justina Ngozi Igwe et al., "Effectiveness of Cognitive Behavior Therapy on Occupational Stress Management Among Administrative, Language, Science and Vocational Education Staff Within Open and Distance Learning Centers: A Randomized Controlled Trial Evaluation," Medicine 103, no. 9 (2024): e37231, https://pubmed.ncbi.nlm.nih.gov/38428863/.

[17] David H. Barlow et al., "The Unified Protocol for Transdiagnostic Treatment of Emotional Disorders Compared with Diagnosis-Specific Protocols for Anxiety Disorders: A Randomized Clinical Trial," JAMA Psychiatry 74, no. 9 (September 2018): 875–84, https://doi.org/10.1001/jamapsychiatry.2017.2164; Beth Fordham et al., "The Evidence for Cognitive Behavioural Therapy in Any Condition, Population or Context: A Meta-Review of Systematic Reviews and Panoramic Meta-Analysis," Psychological Medicine 51, no. 1 (January 2021): 21–29, https://doi.org/10.1017/S0033291720005292.

of our waking activities). As we progress further along toward the highest arousal end of the spectrum, we enter mild stress, then severe stress, and finally, full-blown panic. When we think about stress on a spectrum of low-to-high arousal, we realize that arousal—and what increases or decreases arousal—is the thing to focus on if we wish to deliberately lessen the stress response.

Where we happen to be on the spectrum of arousal at any given moment reflects the level of activity in our autonomic nervous system. Our autonomic nervous system regulates involuntary functions like our heart rate, digestion, blood pressure, and breathing. It also controls our level of alertness, and thus our ability to focus. It is comprised of two sets (sometimes called “arms”) of neural circuits: the parasympathetic nervous system (PNS) and the sympathetic nervous system (SNS). These function more or less independently of one another, but their balance of activities combines to determine our arousal level, and thus our overall brain and bodily state.[18]

But what sets the balance between these two systems? Or, to put it another way, what regulates our autonomic nervous system? There are three main categories of autonomic

[18] J. N. Langley, The Autonomic Nervous System, Part I (W. Heffer & Sons, 1921); Wilfrid Jänig, The Integrative Action of the Autonomic Nervous System: Neurobiology of Homeostasis (Cambridge, UK: Cambridge University Press, 2008).

regulators. The first is our twenty-four-hour circadian rhythm.[19] If you follow a typical diurnal schedule—awake in the daytime and asleep for a major portion of the night—then your SNS activity levels will dominate in the daylight hours and your PNS will dominate at night. In fact, the incredible impact your circadian rhythm has on your ability to reason,[20] exert physical effort,[21] and resist infections[22] stems from how it controls the balance of SNS versus PNS activity. In the morning and early daytime, elevated SNS activity (sometimes called "sympathetic tone") caused by a natural and healthy increase in our baseline

[19] Francesca Baschieri and Pietro Cortelli, "Circadian Rhythms of Cardiovascular Autonomic Function: Physiology and Clinical Implications in Neurodegenerative Diseases," Autonomic Neuroscience: Basic & Clinical 217 (March 2019): 91–101, https://doi.org/10.1016/j.autneu.2019.01.009.

[20] Ilaria Barone et al., "Synaptic BMAL1 Phosphorylation Controls Circadian Hippocampal Plasticity," Science Advances 9, no. 43 (2023), https://doi.org/10.1126/sciadv.adj1010; Shiyang Xu et al., "Relationship Between Circadian Rhythm and Brain Cognitive Functions," Frontiers of Optoelectronics 14, no. 3 (2021): 278–87, https://doi.org/10.1007/s12200-021-1090-y.

[21] Collin M. Douglas et al., "Time of Day and Muscle Strength: A Circadian Output?," Physiology 36, no. 1 (2021): 44–51, https://doi.org/10.1152/physiol.00030.2020; Louis Sebastian Gifford, "Circadian Variation in Human Flexibility and Grip Strength," Australian Journal of Physiotherapy 33, no. 1 (1987): 3–9, https://doi.org/10.1016/S0004-9514(14)60579-1.

[22] Aïssatou Bailo Diallo et al., "For Whom the Clock Ticks: Clinical Chronobiology for Infectious Diseases," Frontiers in Immunology 11 (July 8, 2020): 1457, https://doi.org/10.3389/fimmu.2020.01457; Gareth B. Kitchen et al., "The Clock Gene Bmal1 Inhibits Macrophage Motility, Phagocytosis, and Impairs Defense Against Pneumonia," Proceedings of the National Academy of Sciences 117, no. 3 (2020): 1543–51, https://doi.org/10.1073/pnas.1915932117.

levels of cortisol makes us physically stronger and better able to focus, whereas in the later evening and at night, when our baseline cortisol levels are markedly lower, our parasympathetic tone is elevated, and we become mentally and physically weaker, and our immune system becomes far less capable of warding off infections.[23] (Circadian variation in the expression of specific genes and the actions of various molecules—neuromodulators, hormones, peptides, etc.—are what underlie those dramatic morning-versus-nighttime variations in autonomic tone, as discussed in chapter 1.)

The second category of things that control our autonomic nervous system, and thus our level of arousal, are the sensory inputs—or stimuli, as neuroscientists call them—that we experience. For example, our level of physical comfort or pain, how well-fed or hungry we happen to be (not surprisingly, hunger makes us more alert), whether we are in a brightly lit or dim environment, the environment is quiet or loud, et cetera. And, of course, by the presence or absence of psychological stressors. In general, when we are hungry, thirsty, tired, or otherwise in need of basic sustenance, our arousal levels increase and, with that, our propensity to think and move increases. It should be obvious why that is adaptive. (It helps us seek out food, water, etc.)

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[23] Thomas Steckler et al., Stress and the Brain Part 1: The Neurobiology of Stress (Elsevier Sciences, 2005); Thomas Steckler et al., Stress and the Brain Part 2: Integrative and Clinical Aspects (Elsevier Sciences, 2005).

The third category of things that control our autonomic nervous system is pharmacologic. Caffeine and other stimulants are sympathomimetic, meaning they stimulate the SNS and increase arousal.[24] Other drugs promote PNS activation, and thus decrease arousal levels. For example, alcohol and barbiturates indirectly increase PNS circuit activity,[25] as does the molecule apigenin, which is found in chamomile tea.[26] By the way, if you're wondering why alcohol initially makes people more talkative and alert before it makes them drowsy or has them pass out, it's because before it reduces central nervous system activity and lowers arousal, alcohol first transiently stimulates the SNS. "It lifts you up before it drops you," as they say.

Autonomic means "automatic" and indeed our sympathetic and parasympathetic nervous systems are, for the most part, not under our direct conscious control. We of course can adjust them indirectly. We just discussed three categories of ways to do that. The pathways and mechanisms by which

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[24] Neal L. Benowitz et al., "Sympathomimetic Effects of Paraxanthine and Caffeine in Humans," Clinical Pharmacology & Therapeutics 58, no. 6 (1995): 684–91, https://doi.org/10.1016/0009-9236(95)90025-X.

[25] Ilse C. Schrieks et al., "The Biphasic Effects of Moderate Alcohol Consumption with a Meal on Ambiance-Induced Mood and Autonomic Nervous System Balance: A Randomized Crossover Trial," PLOS One 9, no. 1 (2014): e86199, https://doi.org/10.1371/journal.pone.0086199.

[26] Daniel J. Kramer and Adiv A. Johnson, "Apigenin: A Natural Molecule at the Intersection of Sleep and Aging," Frontiers in Nutrition 11 (February 26, 2024): 1359176, https://doi.org/10.3389/fnut.2024.1359176.

different activities like exercise,[27] or mechanical inputs like massage or a warm bath,[28] or drugs like caffeine[29] elevate or lower arousal by adjusting the activity of our SNS or PNS are fairly well known, but again, it's crucial to note these are all indirect ways of adjusting autonomic arousal. As such, they tend to be slow and hard to titrate—meaning hard to get just right. Moreover, some, like barbiturates or prescription stimulants, have addictive potential and/or are costly.

There is, however, one major exception to all of this: one lever by which we can quickly, directly, and reliably change our level of autonomic arousal by shifting the balance of SNS versus PNS activity. That lever is our respiration pattern, aka our breathing.

Why? Because the neural circuits that control our breathing are part of our autonomic nervous system. In fact, they are the one (and only!) component of it that is both

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[27] Florent Besnier et al., "Exercise Training-Induced Modification in Autonomic Nervous System: An Update for Cardiac Patients," Annals of Physical and Rehabilitation Medicine 60, no. 1 (2017): 27–35, https://doi.org/10.1016/j.rehab.2016.07.002.

[28] Maria Meier et al., "Standardized Massage Interventions as Protocols for the Induction of Psychophysiological Relaxation in the Laboratory: A Block Randomized, Controlled Trial," Scientific Reports 10, article no. 14774 (2020), https://doi.org/10.1038/s41598-020-71173-w; Jian Cui et al., "Repeated Warm Water Baths Decrease Sympathetic Activity in Humans," Journal of Applied Physiology 133, no. 1 (2022): 234–45, https://doi.org/10.1152/japplphysiol.00684.2021.

[29] Brian Fiani et al., "The Neurophysiology of Caffeine as a Central Nervous System Stimulant and the Resultant Effects on Cognitive Function," Cureus 13, no. 5 (2021): e15032, https://doi.org/10.7759/cureus.15032.

autonomic—breathing works just fine without us having to think about it—but that we can also consciously and directly control at any moment we choose. The same cannot be said about our heart rate, digestion, or immune system. As my colleague at Stanford, David Spiegel, MD, has described it to me, breathing and the neural centers that control breathing represent a bridge between our conscious and unconscious mind. We will return to this idea shortly, but to understand the specific patterns of breathing that allow us to consciously and directly control our stress response, and to understand the best tools for adjusting our stress threshold, we need to take the autonomic spectrum—that continuum spanning from deep states of calm to intense states of panic I described earlier—and turn it into a seesaw.

A seesaw? Correct. The simplest model for understanding how to control stress is a seesaw: parasympathetic at one end and sympathetic at the other. When the seesaw is tilted toward the parasympathetic end, we feel calm and relaxed. Tilt it further in that direction and we start drifting off to sleep. A flat, balanced seesaw represents when we are alert, but calm. Tilt it a bit in the other direction, toward the sympathetic end, and we feel more alert and energized. Tilt it further, and we feel stressed. At its steepest tilt, we are in full-blown panic. Reducing the intensity or duration of the stress response can be accomplished either by increasing the load on the parasympathetic side, by reducing the load on the

sympathetic side, or both. How do you do that? We just need to understand a bit about the neural circuits and physiology of the SNS and PNS. And how breathing impacts each of these.

Because it turns out that inhales and exhales modulate the activity of our autonomic nervous system. Inhales reduce PNS activity, leading to relatively more SNS activity and thus, arousal. Exhales directly increase PNS activity. This means your heart rate is faster during inhales than during exhales—which incidentally is the basis of heart rate variability (HRV). Higher HRV in sleep and throughout your day is better than lower HRV, because it means PNS activity is periodically slowing your heart rate down. It's a good marker of your body controlling its own level of arousal, and keeping stress in check.[30] More on all that later but for now, it's going to be helpful to understand what "feeling stressed" is about at a neurochemical level. If you are stressed, that likely means that adrenaline and cortisol have been released and are circulating in your body at higher-than-typical levels, and that norepinephrine has been

[30] Steckler et al., Stress and the Brain Part 1; Steckler et al., Stress and the Brain Part 2; Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, "Heart Rate Variability: Standards of Measurement, Physiological Interpretation and Clinical Use," Circulation 93, no. 5 (March 1996): 1043–65, PMID: 8598068.

released at higher-than-typical levels.[31] You can't just make those molecules go away with positive self-talk. Also, when stress hits, your prefrontal cortex changes its pattern of electrical firing in a way that limits the range of solutions you can come up with to a problem.[32] It even changes your ability to link previous experiences in insightful ways.[33] Most of us are familiar with this stress-induced black-and-white thinking. These facts should help alleviate some of the frustration you may experience when someone tells you to "just relax," and you can't. Or when someone says, "Don't worry, this too shall pass." The first form of encouragement ignores that we must actively do something to our physiology to relax. You can't out-think adrenaline. The latter encouragement is spot on, as stress does indeed hinder our ability to see the big picture, but it's not particularly helpful in the moment. It's sort of like telling you that the splitting headache you're suffering from right

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[31] Georgina Russell and Stafford Lightman, "The Human Stress Response," Nature Reviews Endocrinology 15 (June 2019): 525–34, https://doi.org/10.1038/s41574-019-0228-0.

[32] Amy F. T. Arnsten et al., "The Effects of Stress Exposure on Prefrontal Cortex: Translating Basic Research into Successful Treatments for Post-Traumatic Stress Disorder," Neurobiology of Stress 1 (January 2015): 89–99, https://doi.org/10.1016/j.ynstr.2014.10.002; Elizabeth Woo et al., "Chronic Stress Weakens Connectivity in the Prefrontal Cortex: Architectural and Molecular Changes," Chronic Stress 5 (August 2021), https://doi.org/10.1177/24705470211029254.

[33] Kai A. Schüren et al., "Stress Disrupts Hippocampal Integration of Overlapping Events and Memory Inference in Humans," Science Advances 12, no. 21 (May 2026): eaea5496, https://doi.org/10.1126/sciadv.aea5496.

now won't suck in the future. Fortunately, there is a way to get right to the physiology of the stress response. Rather than try to directly reduce sympathetic tone, you need to instead tilt the seesaw back toward the parasympathetic side by increasing the activity of your PNS. That is going to be the fastest way to calm down in real time whenever you are feeling stressed.

Resilience defined

If you think about it, this is a big part of what resilience is really about. It's not our ability to tolerate or eliminate the stress response but rather our ability to reduce its intensity quickly and in real time, when our arousal exceeds a useful level, so that we can move forward. Resilience is our ability to control stress. Sometimes that means going from extreme stress to very alert and energized, so we can perform a task and quickly problem-solve under intense pressure. Other times it means taking ourselves from endlessly looping thoughts to deep sleep. Resilience is also about an ability to deliberately increase our stress threshold, so life's minor stressors don't disproportionally hijack our mind and body. Once you learn how to do those two things, then you will truly be in the driver's seat.

Maybe that all sounds overly simplistic. Easy to say but harder to do. Well, it turns out there are science-supported tools to adjust your stress response in real time and to raise

your stress threshold, and they work exceedingly well—and fast. They can also work the first time, and every time, because they are grounded in biological mechanisms and circuits that you already have inside you and can access. Some involve adopting a specific pattern of breathing, but they are not breathwork. After all, when we find ourselves in a high stress situation, we can't run off and do five minutes of box breathing! Other protocols don't involve breathing at all. I've been blessed to play a partial role in the discovery and application of several of these stress control protocols, although like so many other scientific "discoveries," aspects of them were known about and used by nonscientists in other contexts first. None of us gets the patent. The following pages explain what they are, how they work, and how to use them.



STRESS CONTROL PROTOCOL

Do One to Three Physiological Sighs to Reduce Stress Fast

WHAT TO DO

- Breathe in the pattern known as the physiological sigh. Inhale deeply through your nose until your lungs are full (your belly should expand out while you inhale), then sharply inhale a second time, also through your nose, to maximally inflate your lungs. Then slowly exhale

through your mouth until your lungs are empty of air. Repeat this in-in-out breathing pattern up to three times, in succession, as needed to reduce your stress level (meaning, your level of autonomic arousal) fast.

HOW IT WORKS

Physiological sighs are, to my knowledge, the fastest and most effective way there is to deliberately calm yourself down. Often just one physiological sigh does the trick.

That's because our state of mind and body changes our breathing but our breathing also changes our state of mind and body.[34] Remember the locus coeruleus that releases norepinephrine to make us alert? Well, specific populations of neurons involved in specific patterns of breathing connect directly to and modulate this release from the locus coeruleus.[35]

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[34] Andrea Zaccaro et al., "How Breath-Control Can Change Your Life: A Systematic Review on Psycho-Physiological Correlates of Slow Breathing," Frontiers in Human Neuroscience 12 (September 2018): 353, https://doi.org/10.3389/fnhum.2018.00353.

[35] Balban et al., "Brief Structured Respiration Practices"; Christina Zelano et al., "Nasal Respiration Entrains Human Limbic Oscillations and Modulates Cognitive Function," Journal of Neuroscience 36, no. 49 (December 2016): 12448–67, https://doi.org/10.1523/JNEUROSCI.2586-16.2016; Peng Li et al., "The Peptidergic Control Circuit for Sighing," Nature 530 (February 2016): 293–97, https://doi.org/10.1038/nature16964; Loren Toussaint et al., "Effectiveness of Progressive Muscle Relaxation, Deep Breathing, and Guided Imagery in Promoting Psychological and Physiological States of Relaxation," Evidence-Based Complementary and Alternative Medicine 2021 (2021): 5924040, https://doi.org/10.1155/2021/5924040; Kevin Yackle et al., "Breathing Control Center Neurons That Promote Arousal in Mice," Science

The conversation between the neurons in your brainstem that control breathing also extends downward, into your spinal cord, which contains neurons that in turn control the muscles between your ribs (called the intercostals) and a dome-shaped muscle attached to the bottom of your rib cage called the diaphragm that actively inflates and deflates your lungs.[36]

When we inhale, oxygen enters our body, fills our lungs, then passes from our lungs into our bloodstream. That only happens because our lungs are not just two big, hollow bags; they contain hundreds of millions of tiny sacs called alveoli, which inflate with air as we inhale. The walls of the alveoli are extremely thin—about 1/2000th of a millimeter (!)—and yet rich with capillaries, the smallest diameter blood vessels. Together, they allow oxygen to move from the air-filled alveoli into the bloodstream. Having hundreds of millions of alveoli greatly increases the surface area through which oxygen can pass into our blood, and thus increases the amount of oxygen we absorb with each breath. (As wild as it sounds, you have approximately 70 to 100 square meters—about the size of a regulation tennis court—of

335, no. 6332 (March 2017): 1411–15, https://doi.org/10.1126/science.aai7984; Sufyan Ashhad et al., “Breathing Rhythm and Pattern and Their Influence on Emotion,” Annual Review of Neuroscience 45 (July 2022): 223–47, https://doi.org/10.1146/annurev-neuro-090121-014424.

[36] Joshua E. Brinkman et al., “Physiology, Respiratory Drive,” StatPearls, last updated June 5, 2023, https://www.ncbi.nlm.nih.gov/books/NBK482414/.

alveolar surface area packed inside your lungs.) Big, deep breaths inflate more alveoli and the more alveoli that inflate, the more oxygen passes into your bloodstream to get delivered to your organs, including your brain.[37]

Carbon dioxide from the bloodstream passes into the alveoli—again through capillaries—and then is expelled from your body when you exhale. We need both oxygen and carbon dioxide, but we need them in the right balance. If we breathe too little, carbon dioxide builds up in our brain and bloodstream, leading to headaches and tiredness.[38] If we breathe too much—rapidly inhaling and exhaling, or what we typically call hyperventilation—we expel too much carbon dioxide, which is problematic because carbon dioxide is crucial for making sure oxygen gets to our brain cells. If you've ever wondered why it's advised to breathe into a paper bag if you're having a panic attack, it's to recapture some of the carbon dioxide that would otherwise be expelled. In this chapter you'll learn faster and better ways to calm down, but the logic is the same. Overbreathing causes

[37] Jänig, Integrative Action; Robert Fried, The Psychology and Physiology of Breathing (Springer, 1993).

[38] Michael Drechsler and Jason Morris, "Carbon Dioxide Narcosis," StatPearls, last updated January 9, 2023, https://www.ncbi.nlm.nih.gov/books/NBK551620/.

anxiety, dizziness, confusion, poor sleep, and even muscle spasms.[39]

As mentioned in the introduction to this chapter, the neurons in our brainstem that control our breathing represent a bridge between our conscious and unconscious mind. They are equipped to sense the ratio of oxygen to carbon dioxide in our blood, which prompts any necessary alterations in our breathing pattern to keep us alive and well. That happens regardless of whether we are exercising hard or deeply relaxed—it even happens while we sleep.[40] Unlike other autonomic neural circuitry, however, we can take immediate conscious control of these neurons. We can decide "I'm going to extend my next exhale a bit" or "I'm going to take one more inhale before holding my breath" and—voila!—we can make that happen. We just don't have that level of conscious control over any other autonomic function. There are some reported examples of people who used biofeedback to learn to directly control their heart rate or the size of the pupils of their eyes, but that requires a fair amount of

[39] C. Gilbert, "Hyperventilation and the Body," Accident and Emergency Nursing 7, no. 3 (1999): 130–40, https://doi.org/10.1016/s0965-2302(99)80072-1.

[40] E. Nattie, "CO2, Brainstem Chemoreceptors and Breathing," Progress in Neurobiology 59, no. 4 (1999): 299–331, https://doi.org/10.1016/s0301-0082(99)00008-8.

training.[41] Extended exhales and physiological sighs are something that anyone can do.

Certain internal states are associated with certain respiratory patterns. And by deliberately changing how we breathe, we can shift our internal state. That's a big deal and something I wish they had taught us in grade school. Alas, it wasn't until recently that Western science embraced controlled studies of respiration and, in doing so, revealed what had been known in Eastern traditions for thousands of years. We owe it to a small set of dedicated researchers, mostly neuroscientist Jack Feldman's lab at UCLA, for discovering how the brain controls breathing, both our unconscious reflexive breathing and deliberate breathing, and how they impact our emotional and cognitive states.[42] Before and during our discussion on the Huberman Lab podcast, Jack told me stories of setting out to study this topic in the early 2000s, and a luminary of neuroscience making fun of him at meetings. He would jab: "Why would you study something as boring as breathing?

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[41] Sarah Nadine Meissner et al., "Self-Regulating Arousal via Pupil-Based Biofeedback," Nature Human Behaviour 8 (2024): 43–62, https://doi.org/10.1038/s41562-023-01729-z.

[42] Jeffrey C. Smith et al., "Pre-Bötzinger Complex: A Brainstem Region That May Generate Respiratory Rhythm in Mammals," Science 254, no. 5032 (1991): 726–29, https://doi.org/10.1126/science.1683005; Paul A. Gray et al., "Normal Breathing Requires Pre-Bötzinger Complex Neurokinin-1 Receptor-Expressing Neurons," Nature Neuroscience 4 (2001): 927–30, https://doi.org/10.1038/nn0901-927; Li et al., "Peptidergic Control Circuit."

Why don't you study something important, like memory?"[43] (That guy studied memory.) Jack knew to ignore him and follow his intuition. His work ignited a field that is now positively impacting the mental and physical health of people of all ages. Detecting and modifying certain breathing patterns also turns out to be important for predicting heart attacks,[44] strokes,[45] and seizures,[46] and for preventing sudden infant death syndrome (SIDS).[47] Some of his recent work has also shed light on the fact that some opioid drugs can lead to death because of how they change the firing

[43] Andrew Huberman, "Dr. Jack Feldman: Breathing for Mental & Physical Health and Performance," Huberman Lab (podcast audio), November 13, 2025, https://www.hubermanlab.com/episode/essentials-breathing-mental-physical-health-performance-jack-feldman.

[44] Alberto Giannoni et al., "Upright Cheyne-Stokes Respiration in Patients with Heart Failure," JACC 75, no. 23 (2020): 2934–46, https://doi.org/10.1016/j.jacc.2020.04.033.

[45] R. Doug McEvoy et al., "The Sleep Apnea Cardiovascular Endpoints (SAVE) Trial: Rationale and Start-Up Phase," Journal of Thoracic Disease 2, no. 3 (2010): 138–43, https://doi.org/10.3978/j.issn.2072-1439.2010.02.03.5; R. Doug McEvoy et al., "Erratum to the Sleep Apnea Cardiovascular Endpoints (SAVE) Trial," Journal of Thoracic Disease 8, no. 8 (2016): E850, https://doi.org/10.21037/jtd.2016.07.79.

[46] Heidi L. Grabenstatter, "Irregular Respiratory Rhythm: A Physiological Biomarker of SUDEP Risk in Patients with Nocturnal Seizures?," Epilepsy Currents 16, no. 5 (2016): 327–29, https://doi.org/10.5698/1535-7511-16.5.327; Rup K. Sainju et al., "Interictal Respiratory Variability Predicts Severity of Hypoxemia After Generalized Convulsive Seizures," Epilepsia 64, no. 9 (2023): 2373–84, https://doi.org/10.1111/epi.17691.

[47] V. L. Schechtman et al., "Dynamics of Respiratory Patterning in Normal Infants and Infants Who Subsequently Died of the Sudden Infant Death Syndrome," Pediatric Research 40, no. 4 (1996): 571–77, https://doi.org/10.1203/00006450-199610000-00010.

patterns of the neurons that control breathing.[48] It is fair to say that Jack's work has literally saved lives.

Breathing isn't just about getting oxygen into your lungs; it also helps control heart rate. While you can't directly control your heart rate, you can leverage the fact above and use your breathing to control it indirectly. Deliberately inhaling more vigorously or longer (or both) increases your heart rate, whereas deliberately exhaling more vigorously or for longer slows your heart rate down, through a process called respiratory sinus arrhythmia. Exhales also activate a specific branch of the vagus nerve, which is part of the PNS, which in turn reduces the rate of electrical activity in neurons of the sinoatrial node of the heart, your heart's natural pacemaker. So deliberately emphasizing the intensity and/or duration of your exhales for a few breathing cycles will caaaaalm youuuu dooooown. The breathing pattern described in this protocol does just that.

Once you learn the relationship between breathing and heart rate, and deploy the physiological sigh to reduce your heart rate on the spot, it gives you a significant feeling of confidence in dealing with stress, and uncertain circumstances too. You no longer have to be afraid you will get stuck in an anxiety response in a key moment, or any

[48] Xiaolu Sun et al., "Opioids Modulate an Emergent Rhythmogenic Process to Depress Breathing," eLife 8 (2019): e50613, https://doi.org/10.7554/eLife.50613.

moment for that matter. You also realize that what most people say to themselves or others when stress hits—"Take a deep breath"—is the exact opposite of what we need to do! When stress hits we should do a big exhale. It is when we really draw out that exhale that our heart rate starts to slow down.

So why am I recommending "Do one to three physiological sighs to calm down fast" and not just "Do a big exhale"? And why aren't I suggesting box breathing, a popular breathing pattern that asks you to inhale-hold-exhale-hold, each in equal duration? Because some breathing techniques are far more potent and fast acting than others for stress reduction. And physiological sighs are particularly powerful for that purpose, because they capitalize on a hardwired set of neural circuits and neuromuscular pathways, which at once slow our heart rate down and adjust our oxygen–to–carbon-dioxide ratio in our brain and our bloodstream. The first two steps of the physiological sigh—the long inhale, followed by the quick inhale—are vital because of those hundreds of millions of alveoli we discussed. The insides of alveoli are a bit damp, not unlike a deflated balloon that once held water. And they tend to stick together due to surface tension. The long inhale inflates many alveoli, and the second quick inhale that immediately follows helps generate the force needed to reinflate alveoli that would otherwise stay deflated. This is especially

effective when you inhale through your nose, because nasal breathing generates more resistance than mouth breathing and thus more force to reinflate those alveoli. Doing a physiological sigh also stimulates a collateral (meaning a branch of a neural pathway) that emerges from the brainstem area controlling your breathing and reaches other brain areas important for emotional and cognitive control.[49] Thus, physiological sighs, unlike just a big exhale or box breathing, are a three-pronged tool to quickly (1) slow your heart rate, (2) reestablish proper oxygen–to–carbon–dioxide ratios in your brain and bloodstream, and (3) engage brain areas that let you think logically about your situation. It's adaptive biological engineering at its best and works to calm you down.

Infants do physiological sighs reflexively.[50] Animals do too.[51] Observe a dog right before it goes down for a nap. You'll see that double inhale, long exhale pattern. Rats physiological sigh when placed into a claustrophobic

[49] Jan-Marino Ramirez, "The Integrative Role of the Sigh in Psychology, Physiology, Pathology, and Neurobiology," Progress in Brain Research 209 (2014): 91–129, https://doi.org/10.1016/B978-0-444-63274-6.00006-0.

[50] Anna S. Lijowska et al., "Sequential Arousal and Airway-Defensive Behavior of Infants in Asphyxial Sleep Environments," Journal of Applied Physiology 83, no. 1 (1997): 219–28, https://doi.org/10.1152/jappl.1997.83.1.219.

[51] Stefan Soltysik and Piotr Jelen, "In Rats, Sighs Correlate with Relief," Physiology & Behavior 85, no. 5 (2005): 598–602, https://doi.org/10.1016/j.physbeh.2005.06.008.

environment.[52] People reflexively do physiological sighs about once every few minutes when working[53] and even in their sleep to reset their oxygen–to–carbon-dioxide ratio.[54]

In fact, you have dedicated sets of neurons in your brainstem for generating physiological sighs.[55] Nature just doesn't create discrete neural pathways by accident; that's how important this breathing pattern is. One respiratory region, the pre-Bötzinger complex, controls your rhythmic breathing, the regular alternation of inhales and exhales.[56] But a second region, the parafacial nucleus, controls breathing that involves double inhales, double exhales, or any combination of inhales and exhales in ratios other than the 1:1 inhale-exhale pattern.[57] The parafacial nucleus does

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[52] Karolyne S. Magalhães et al., "Locus Coeruleus as a Vigilance Centre for Active Inspiration and Expiration in Rats," Scientific Reports 8, article no. 15654 (2018), https://doi.org/10.1038/s41598-018-34047-w.

[53] Elke Vlemincx et al., "A Sigh Following Sustained Attention and Mental Stress: Effects on Respiratory Variability," Physiology & Behavior 107, no. 1 (2012): 1–6, https://doi.org/10.1016/j.physbeh.2012.05.013.

[54] Rogelio Perez-Padilla et al., "Sighs During Sleep in Adult Humans," Sleep 6, no. 3 (1983): 234–43, https://doi.org/10.1093/sleep/6.3.234.

[55] Liza J. Severs et al., "The Psychophysiology of the Sigh: I: The Sigh from the Physiological Perspective," Biological Psychology 170, article no. 108313 (April 2022), https://doi.org/10.1016/j.biopsycho.2022.108313.

[56] Gray et al., "Normal Breathing Requires Pre-Bötzinger Complex."

[57] Annette Pisanski and Silvia Pagliardini, "The Parafacial Respiratory Group and the Control of Active Expiration," Respiratory Physiology & Neurobiology 265 (July 2019): 153–60, https://doi.org/10.1016/j.resp.2018.06.010; Robert T. R. Huckstepp et al., "Role of Parafacial Nuclei in Control of Breathing in Adult Rats," Journal of Neuroscience 35, no. 3 (2015): 1052–67, https://doi.org/10.1523/JNEUROSCI.2953-14.2015; Li et al., "Peptidergic Control Circuit."

things other than generate physiological sighs but that's one of its primary functions.

I'd like to clarify that physiological sighs are not breathwork. Breathwork refers to setting aside five to twenty minutes to do a specific breathing pattern (or designated sequence of patterns) to get you into a specific state of mind and body. Physiological sighs are a naturally occurring breathing cycle that we can deliberately recreate. And they are the fastest way to consciously tilt your seesaw to the parasympathetic side of your autonomic nervous system.

Do it before public speaking or a hard conversation. Do it when you're stuck in traffic or late to an important meeting and starting to feel your tension level rise. Do it to stop yourself from overreacting to an offhand comment. It won't guarantee that you'll always do the right thing in response, but it will put you in a better frame of body and mind, so that you have the best chance of navigating the stress that comes from these or similar situations.

There are data showing that doing physiological sighs outside of stressful moments can help buffer you against future stress too.[58] Several years ago, my lab collaborated with psychiatrist David Spiegel to investigate just that.[59] We

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[58] Elke Vlemincx et al., "Why Do You Sigh? Sigh Rate During Induced Stress and Relief," Psychophysiology 46, no. 5 (2009): 1005–13, https://doi.org/10.1111/j.1469-8986.2009.00842.x.

[59] Balban et al., "Brief Structured Respiration Practices."

conducted a study that randomized 108 participants to one of four groups. One group did five minutes per day of cyclic physiological sighing: Inhale, inhale, long exhale, repeat, and so on for five minutes. The second group did five minutes per day of box breathing, a breathing technique I've mentioned before, which follows a pattern of inhale-hold-exhale-hold-repeat, where each inhalation and exhalation, and the holds, are the same duration, until five minutes have elapsed. The third group did five minutes per day of cyclic hyperventilation: You take a big deep breath in through the nose and a shorter exhale through the mouth, putting more vigor into the inhalation than the exhalation. A fourth group did five minutes per day of meditation, seated with eyes closed, and focusing their attention on their breath but with no specific instructions about breathing pattern.

Most participants, regardless of their group, reported an improvement in mood and outlook. But cyclic physiological sighing had a notably bigger effect. We also observed improvements in resting heart rate, improved sleep, and other metrics of well-being related to autonomic function. In other words, doing one to three physiological sighs is a great tool for when stress hits and doing five minutes of physiological sighs on repeat can help buffer us further against the inevitable stress of life. Given what you now know about exhales slowing heart rate, oxygen and carbon dioxide levels, and the autonomic seesaw, it should make

sense as to why this works. Your brain and body were literally built to do this.



STRESS CONTROL PROTOCOL

Use Deliberate Cold Exposure to Raise Your Stress Threshold

WHAT TO DO

- Put yourself in uncomfortably cold water three to seven times per week. This can be as simple as a cold shower, or as involved as an ice bath. For the duration, choose one of these options: Timed protocol: One to three minutes per session.
- Count “walls of adrenaline” protocol: Rather than committing to a specific amount of time, try committing to a specific number of mental “walls” to climb over. When you feel the impulse to avoid or exit the cold but you stay, that’s one wall you’ve scaled. Decide in advance how many times you will encounter this impulse and keep going. You do not have to always increase the number of walls. There are days when you will decide to “go over” five walls and other days where two walls will be enough.
- The water should be uncomfortably cold. This means it should immediately trigger your impulse to get out, but it should be warm enough that you can safely stay in the

water for up to three minutes. Why do I suggest using such annoyingly subjective criteria? Because the specifics will differ by person, the time of day, the ambient temperature, how well rested you are, and on and on. It should be cold enough to require you to overcome some mental resistance to get in. And once you get in, it should be cold enough to cause your breathing to quicken. It should make you want to raise your shoulders. If the water is so cold that it burns or gives you a headache, however, it's probably too cold. For most people, somewhere between 40 and 50 degrees Fahrenheit works great.

HOW IT WORKS

The previous protocol will help reduce the intensity and the duration of your stress response. This one takes a different approach to stress control by increasing your stress threshold.

But let's first address some of the common myths about deliberate cold exposure. You may have heard that it can increase your metabolism. In my view, the data are not strong enough to support those claims, at least not by doing a typical one-to-three-minute cold shower or cold immersion (and staying in the cold much longer than that could be unsafe).

You also may have heard to avoid deliberate cold exposure after any resistance training session aimed at increasing muscle growth and/or strength. That counsel is legitimate. Deliberate cold exposure in the six to eight hours after resistance training can prevent some of the strength and hypertrophy adaptations that would otherwise occur. (Chapter 2 explains more about those adaptations, and how to achieve them.) That inhibition of muscle growth is likely due to the vasoconstriction the cold causes, which limits both nutrient delivery and waste removal from the recently trained muscles and also blunts protein synthesis.[60] Wait at least six hours after resistance training to do deliberate cold exposure, or do it before you train, or just save it for another day. Or do it any time relative to your resistance training, but be sure to end with a hot shower or bath to reestablish perfusion of the muscles to correct for this. One study argued that deliberate cold exposure done in the first six hours after resistance training does not prevent strength adaptations, but most of the work on this issue indicates that deliberate cold exposure done in the six hours or so after resistance training will significantly reduce any hypertrophy or strength

[60] Jackson J. Fyfe et al., "Cold Water Immersion Attenuates Anabolic Signaling and Skeletal Muscle Fiber Hypertrophy, but Not Strength Gain, Following Whole-Body Resistance Training," Journal of Applied Physiology 127, no. 5 (2019): 1403–18, https://doi.org/10.1152/japplphysiol.00127.2019.

increases that would otherwise occur.[61] So it is best to simply do it before resistance training, or on rest days. Or allow six or more hours to elapse after resistance training before deliberate cold exposure.

Another common myth is that deliberate cold exposure is harmful for people wishing to limit their cortisol levels. You often hear that people should avoid deliberate cold exposure to keep their cortisol levels low too. Yet the cortisol increases caused by deliberate cold exposure, while significant, are transient and small compared to the dramatic increase that a moderate intensity exercise session triggers.[62] You won't hear that people should avoid exercise to keep

[61] Fyfe et al., "Cold Water Immersion Attenuates"; Elvis S. Malta et al., "The Effects of Regular Cold-Water Immersion Use on Training-Induced Changes in Strength and Endurance Performance: A Systematic Review with Meta-Analysis," Sports Medicine 51, no. 1 (January 2021): 161–74, https://doi.org/10.1007/s40279-020-01362-0; Llion A. Roberts et al., "Post-Exercise Cold Water Immersion Attenuates Acute Anabolic Signalling and Long-Term Adaptations in Muscle to Strength Training," Journal of Physiology 593, no. 18 (September 2015): 4285–301, https://doi.org/10.1113/JP270570; Jozo Grgic, "Effects of Post-Exercise Cold-Water Immersion on Resistance Training–Induced Gains in Muscular Strength: A Meta-Analysis," European Journal of Sport Science 23, no. 3 (March 2023): 372–80; Alec Piñero et al., "Throwing Cold Water on Muscle Growth: A Systematic Review with Meta-Analysis of the Effects of Postexercise Cold Water Immersion on Resistance Training–Induced Hypertrophy," European Journal of Sport Science 24, no. 2 (February 2024): 177–89.

[62] Milda Eimonte et al., "Residual Effects of Short-Term Whole-Body Cold-Water Immersion on the Cytokine Profile, White Blood Cell Count, and Blood Markers of Stress," International Journal of Hyperthermia 38, no. 1 (2021): 696–707, https://doi.org/10.1080/02656736.2021.1915504; E. E. Hill et al., "Exercise and Circulating Cortisol Levels: The Intensity Threshold Effect," Journal of Endocrinological Investigation 31, no. 7 (2008): 587–91, https://doi.org/10.1007/BF03345606.

cortisol low because the cortisol increase caused by exercise is, again, transient and, what's more, beneficial. (Indeed, it is a key aspect of the specific and general adaptation triggers we discussed in chapter 2).

Many of the "cold exposure is bad for you" stories involve people going into the cold for very long periods of time at excruciatingly cold temperatures, or combining deliberate cold exposure with hyperventilation breathwork and/or jumping or diving under ice. All of which are dangerous and should be avoided.

Now that those points are out of the way, let's look at why deliberate cold exposure is beneficial. Done correctly, deliberate cold exposure is a powerful, safe, and reliable tool for rapidly adjusting your internal state and stress threshold. There are two key phases, each with distinct benefits. First, the state shift that occurs during the cold exposure itself. Second, the state shift that occurs for many hours afterward. Anyone who has done deliberate cold exposure consistently—that is, three to seven times per week—will tell you there is a third phase too, which is the change in your subjective experience of stress and your ability to navigate other forms of stress.

When you enter the cold, it sharply tilts your autonomic nervous system toward the sympathetic end of the seesaw. It does this the first time and every time, and it happens fast,

within one to two seconds upon entering the cold. Then, once you get out of the cold, you tend to remain in a very "alert but calm" state—meaning (in the context of the model I described) with an evenly balanced autonomic seesaw—for many hours afterward.

Deliberate cold exposure increases the levels of circulating catecholamines in your brain and body: adrenaline, norepinephrine, and dopamine, and increases in the hormone cortisol too.[63] Some, including dopamine in particular, remain elevated long after you dry off and warm up.[64] These catecholamines increase mood, energy, and alertness. They also can increase your stress threshold; more on that below.[65] But first I want to pause for a moment because you may be asking this: Why would I want to trigger the release of these "stress" chemicals? Won't that make me feel more, not less, stressed? Yes, but that's what this protocol is all about.

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[63] B. P. O'Malley et al., "Circulating Catecholamine, Thyrotrophin, Thyroid Hormone and Prolactin Responses of Normal Subjects to Acute Cold Exposure," Clinical Endocrinology 21, no. 3 (1984): 285–91, https://doi.org/10.1111/j.1365-2265.1984.tb03471.x.

[64] P. Šrámek et al., "Human Physiological Responses to Immersion into Water of Different Temperatures," European Journal of Applied Physiology 81 (February 2000): 436–42, https://doi.org/10.1007/s004210050065.

[65] C. P. Arun, "Fight or Flight, Forbearance and Fortitude: The Spectrum of Actions of the Catecholamines and Their Cousins," Annals of the New York Academy of Sciences 1018, no. 1 (2004): 137–40, https://doi.org/10.1196/annals.1296.016; Akil, "Stress Fitness."

We need to challenge the notion that stress is always bad for us.

In fact, when we have control over when, how much, and especially for what purpose we experience stress, the impact of that stress can switch from being health-perturbing to health-promoting, and this reshapes the body's response to all forms of stress.[66] This is not "mind over matter," it's leveraging the brain circuits that convert the arousal response we call stress into something called eustress—a term coined by Swiss physiologist Hans Selye.[67] (You may remember him from the previous chapter.) He was nominated for the Nobel Prize seventeen times for his work on this and related topics,[68] long before we knew what the underlying mechanisms were, but as those mechanisms have become elucidated, many of his theories have largely been deemed correct.

We now know that your prefrontal cortex—that patch of brain real estate involved in context-dependent decision-making (as well as "top-down" inhibition of impulsive actions) is also a key component of our limbic system, which is involved in different arousal-based

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66 Crum et al., "Rethinking Stress."

67 Hans Selye, Stress Without Distress (Signet, 1974).

68 "Hans Selye," Nomination Archive, Nobel Prize, n.d., https://www.nobelprize.org/nomination/archive/show_people.php?id=8395.

responses.[69] Usually when people talk about the limbic system, they focus on the amygdala, a brain structure often touted as our fear center. They also talk about our reptilian brain, which is pop neuroscience language for structures like our hypothalamus, which contains dozens of subareas involved in regulating hunger, thirst, aggression, mating, and much more. The hypothalamus and amygdala are often placed conceptually opposite "more evolved" or "higher" brain centers like the prefrontal cortex. But alas, that framework is a gross misrepresentation of the way our brain actually works.

Human brains don't have one system for positive arousal and another entirely different system for negative arousal. We simply have a system for autonomic arousal that is linked to, and impacted by, areas of the brain that evaluate the context of our experiences and then help us make adaptive decisions. Think about it: There are an infinite number of different contexts in which arousal could be good, and just as many where it could be bad. Your prefrontal cortex carries out the basic analysis when arousal happens, asking questions like, "Did I choose this situation?" "Can I control how long and/or how intense this arousal is?" and most importantly, "Is this state of elevated autonomic arousal good or bad for

[69] Ned H. Kalin, "Prefrontal Cortical and Limbic Circuit Alterations in Psychopathology," American Journal of Psychiatry 176, no. 12 (2019): 971–73, https://doi.org/10.1176/appi.ajp.2019.19101036.

me?" Of course, the answers to those questions depend on the context. And your prior experiences and knowledge.

In fact, our knowledge about stress—not just our memories of stressful circumstances we have been in before, but what we believe to be true about stress—affects how we respond to it. Work from Alia Crum's lab at Stanford shows that humans who learn about some of the (true) positive effects of stress-related arousal on physical health and mental performance experience those positive effects when they are subjected to stress.[70] Her lab also showed that if people learn a bunch of (also true) information about how stress can make us physically unhealthy and cognitively impaired, well, that's what happens to them when they get stressed![71] We can thank our prefrontal cortex for that ability to reframe stress in ways that change its actual effects on us. It's not just a decrease in its negative effect but an inversion from deleterious to beneficial. There's more that shapes our experience of stress as good or bad. Perhaps one of the most interesting things that determines the outcome is whether the stressor has a known end point and resolution that we control.[72] With that in mind, it should now make perfect

[70] Crum et al., "Evaluation of the 'Rethink Stress.' "

[71] Crum et al., "Rethinking Stress"; Crum et al., "Role of Stress Mindset"; Crum et al., "Optimizing Stress"; Crum et al., "Evaluation of the 'Rethink Stress.' "

[72] Linghui Yang et al., "Effects of Stressor Predictability and Controllability on Sleep, Temperature, and Fear Behavior in Mice," Sleep 34, no. 6 (2011): 759–71, https://doi.org/10.5665/SLEEP.1044; Chirag Limbachia et al., "Controllability over Stressor Decreases Responses in Key Threat-Related

sense why deliberate cold exposure meets all the criteria of eustress, and why I coined it deliberate. If you are the one who elected it, and you control its duration, and you understand how it is good for you, you can turn stress into eustress.

So how, exactly, is it good for you? Two ways. First, the stress of deliberate cold exposure helps us become better at handling other moments of stress. That's sometimes called cross adaptation. It's often studied in the context of physical stress, where exposure to one type of stress—say, exercising in extreme heat—prepares a person to tolerate another type of stress, such as thriving in environments with low oxygen levels.[73] The idea is that we can precondition ourselves to cope with challenges by seeking out situations that draw on similar aspects of our physiology. Cold water swimming has been proposed as a way to improve surgical outcomes,[74] for example, because the stress of the former helps patients better handle the stress of the latter.

Brain Areas," Communications Biology 4, article no. 42 (2021), https://doi.org/10.1038/s42003-020-01537-5; Dakota D. Witzel et al., "Daily Association Between Perceived Control and Resolution of Daily Stressors Strengthens Across a Decade of Adulthood," Communications Psychology 3, article no. 130 (2025), https://doi.org/10.1038/s44271-025-00313-7.

73 Yuval Heled et al., "Heat Acclimation and Performance in Hypoxic Conditions," Aviation, Space, and Environmental Medicine 83, no. 7 (2012): 649–53, https://doi.org/10.3357/asem.3241.2012.

74 C. Mark Harper, "Extreme Preconditioning: Cold Adaptation Through Sea Swimming as a Means to Improving Surgical Outcomes," Medical Hypotheses 78, no. 4 (2012): 516–19, https://doi.org/10.1016/j.mehy.2012.01.022.

Controlled studies showing that cross adaptation focused on physical stress can prepare us to face emotional stress are slowly accruing. In one study of eighty-six people, combining cold exposure with breathing techniques changed the participants' perception of stress in their everyday lives.[75] Participants rated statements such as "You feel calm" and "You enjoy yourself" following a few minutes of vigorous breathwork and a quick cold shower every morning for a period of two weeks. And participants who did both during that period reported a greater sense of control over irritations, and stronger feelings that life was going their way and that they were on top of things, compared to participants who did only the breathwork or the cold exposure, or neither.

Some data are more tangential in nature. For example, cold exposure appears to improve the body's response over time to hypoxia, or low oxygen levels in the blood, which is relevant because hypoxia can rapidly set in when we hit a state of anxiety or panic.[76] Some preliminary evidence has pointed to a possible role for cold exposure in the treatment of depression, not as a replacement for medication but as an

[75] Cristopher Siegfried Kopplin and Louisa Rosenthal, "The Positive Effects of Combined Breathing Techniques and Cold Exposure on Perceived Stress: A Randomised Trial," Current Psychology 42 (October 2022): 27058–70, https://doi.org/10.1007/s12144-022-03739-y.

[76] Heather C. Lunt et al., " 'Cross-Adaptation': Habituation to Short Repeated Cold-Water Immersions Affects the Response to Acute Hypoxia in Humans, Journal of Physiology 588, pt. 18 (July 2010): 3605–13,

accompaniment.[77] In 2018, a group of researchers from the UK reported that a twenty-four-year-old woman who had been treated for major depressive disorder and anxiety since the age of seventeen with little to no relief began weekly sessions of cold water swimming in an attempt to find something that could diminish, or negate, the need for medication after she became a mother. The intervention worked for her. Within four months, she no longer required drug treatment.[78] That was a case study of just one person, but interesting nonetheless.

A 2023 review of fourteen studies on the mental health benefits of outdoor cold-water swimming found this practice to be associated with stress reduction, mood improvement, and a positive outlook.[79] That result is not necessarily

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https://doi.org/10.1113/jphysiol.2010.193458; Lyudamila T. Kovtun and Mikhail I. Voevoda, Susceptibility to Hypoxia and Breathing Control Changes After Short-Term Cold Exposures," International Journal of Circumpolar Health (August 2013): 72, https://doi.org/10.3402/ijch.v72i0.21574; Hitoshi Wakabayashi et al., "Recent Updates on Cold Adaptation in Population and Laboratory Studies, Including Cross-Adaptation With Nonthermal Factors," Journal of Physiological Anthropology 44 (February 2025): 7, https://doi.org/10.1186/s40101-025-00387-6.

77 Nikolai A. Shevchuk, "Adapted Cold Shower as a Potential Treatment for Depression," Medical Hypotheses 70, no. 5 (2008): 995–1001, https://doi.org/10.1016/j.mehy.2007.04.052.

78 Christoffer van Tulleken et al., "Open Water Swimming as a Treatment for Major Depressive Disorder," BMJ Case Reports 2018 (August 21, 2018): bcr2018225007, https://doi.org/10.1136/bcr-2018-225007.

79 K. Overbury et al., "Swimming in Nature: A Scoping Review of the Mental Health and Wellbeing Benefits of Open Water Swimming," Journal of Environmental Psychology 90 (September 2023): 102073, https://doi.org/10.1016/j.jenvp.2023.102073.

entirely tied to cold; being outdoors and part of a community (many people do winter swims in groups or with friends) certainly contributes, but cold exposure is responsible for at least some of this benefit. Other work has also shown a connection between winter swimming and well-being.[80] In one study, fifty-three people in the UK who participated in eight weekly sea swims reported a reduction in both depression and anxiety.[81] Interestingly, "confronting a challenge" was commonly reported as a reason why the experience was rewarding. In another investigation, forty-two undergraduates at the University of Chichester, in the UK, completed a mood profile questionnaire before and after cold immersion, with an additional twenty-two students completing the questionnaire twice but without cold immersion in between.[82] Energy and self-esteem increased significantly in the experimental group, which also experienced a decrease in tension, anger, and depression. No comparable changes were observed in the control group.

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[80] Pirkko Huttunen et al., "Winter Swimming Improves General Well-Being," International Journal of Circumpolar Health 63, no. 2 (2004): 140–44, https://doi.org/10.3402/ijch.v63i2.17700.

[81] Amy Burlingham et al., "Sea Swimming as a Novel Intervention for Depression and Anxiety—a Feasibility Study Exploring Engagement and Acceptability," Mental Health and Physical Activity 23 (October 2022), https://doi.org/10.1016/j.mhpa.2022.100472.

[82] John S. Kelly and Ellis Bird, "Improved Mood Following a Single Immersion in Cold Water," Lifestyle Medicine 3, no. 1 (2022): e53, https://doi.org/10.1002/lim2.53.

The overarching concept here is really about the intentional discomfort brought about from doing deliberate cold exposure. As Mike Tipton, a sports and exercise scientist who has studied cold exposure extensively at the University of Portsmouth, has noted, the kind of discomfort stemming from a cold bath was once a commonplace occurrence.[83] Humans have been exposed to cold for a very long time, and modern living has eliminated it. As a result, we are never forced to move out of homeostasis, which means our various systems are rarely challenged. "The dynamic equilibrium that underpins homeostasis needs to be perturbed," Tipton wrote in an editorial first published in 2018. Stanford University psychiatrist Anna Lembke, author of Dopamine Nation, has also spoken about the role of deliberate pain exposure to combat addictions. As she explained it, we are living in the midst of a "fire hose of dopamine," triggered by our phones, caffeine, sugar, and other aspects of modern life.[84] And part of the way out of this is to temporarily abstain from pleasure and to intentionally seek pain—not life-threatening pain, not anything dangerous, but

[83] Mike Tipton, "Humans: A Homeothermic Animal That Needs Perturbation?," Experimental Physiology 104, no. 1 (2019): 1–2, https://doi.org/10.1113/EP087450.

[84] Anne Lembke, Dopamine Nation: Finding Balance in the Age of Indulgence (Penguin, 2021).

discomfort.[85] We need to learn to cope with stress, not just eliminate it. Stress is a part of life, and we are capable of withstanding it. Cold immersion shows us how.

The second way deliberate cold exposure is good for us is the state it puts us in after it's over. And no, I'm not just talking about a state of relief—although it is fair to say that nearly everyone enjoys getting out of the cold plunge or cold shower more than getting into it. In all seriousness, very few people actually enjoy deliberate cold exposure while they are in the cold. It is a potent stressor and no amount of learning about eustress will change that. In fact, I often find myself cursing Selye when I'm in the cold. But once we get out, ahhh, that's when the real magic starts. The sharp increase in adrenaline, norepinephrine, and cortisol we just triggered also sets in motion a dramatic and prolonged increase in dopamine—the molecule that, as we will discuss more in chapter 6, induces a heightened sense of motivation and focus and, with the pain of the cold now gone, buzz. Dopamine doesn't act alone to create the post-deliberate-cold state, but it is a central player in that neuroendocrine cocktail. It's the perfect state for a work session, for exercising, socializing, pretty much any activity that requires us to be alert and focused.

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[85] "Too Much Pleasure Can Lead to Addiction. How to Break the Cycle and Find Balance," Life Kit, NPR, April 4, 2022, https://www.npr.org/transcripts/1090009509.

If the way I'm describing the post-cold state sounds almost drug-like, it is. Only better. One study found that full-body (up to the neck) immersion in water at about 57 degrees Fahrenheit for one hour triggered a more than fivefold increase in norepinephrine levels compared to baseline.[86] It also produced a rise in dopamine levels in the body, raising them more than twofold for hours afterward. We call this peripheral dopamine, because it was measured from the body. We don't yet know the exact relationship between dopamine levels in the body and in the brain, but we do know they tend to increase and decrease in parallel. And there is emerging evidence that when peripheral dopamine is elevated, it increases dopamine in reward and motivation circuits in the brain in a dose-dependent manner.[87] When adrenaline is released in the body, it doesn't cross the blood-brain barrier but it does activate receptors in the vagus nerve that project from the body to the brain. In doing so, it causes the locus coeruleus to release norepinephrine, which increases alertness in the brain. So there is reason to assume that peripheral increases in catecholamines triggered by

[86] Šrámek et al., "Human Physiological Responses."

[87] J. Daniel Obray et al., "Dopamine D2-Subtype Receptors Outside the Blood-Brain Barrier Mediate Enhancement of Mesolimbic Dopamine Release and Conditioned Place Preference by Intravenous Dopamine," Frontiers in Cellular Neuroscience 16 (2022): 944243, https://doi.org/10.3389/fncel.2022.944243; G. Heninger et al., "Plasma and CSF Dopamine Metabolites in Psychiatric Patients: Effects of Probenecid and Haloperidol," Catecholamines: Basic and Clinical Frontiers (1979): 1887–89, https://doi.org/10.1016/B978-1-4832-8363-0.50580-2.

deliberate cold exposure are mirrored by similar increases in the brain.

It is worth noting that drugs known to sharply increase dopamine and epinephrine, such as amphetamines or cocaine, also produce fast, large increases in catecholamines but with the important caveat that they cause levels of the catecholamines to rise so fast and then drop so fast, that they plummet below pre-drug levels. That is part of what, as Anna Lembke, MD, explained in Dopamine Nation, puts people in a state of psychological and physical discomfort, and leaves them craving more of those drugs right away, which partially explains why they are addictive.[88]

Deliberate cold exposure then seems an almost perfect way to significantly increase and maintain elevated levels of catecholamines. You get the mental focus and mood benefits they provide, and you also raise your stress threshold.

Most of us don't have an hour available to spend sitting in chilly water, however. Fortunately, you can use colder temperatures for much shorter periods of time to get the same effect. Studies show that even thirty seconds can work.[89] Yet I suggest one to three minutes or two to ten

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[88] Lembke, Dopamine Nation.

[89] Geert A. Buijze et al., "The Effect of Cold Showering on Health and Work: A Randomized Controlled Trial," PLOS One 11, no. 9 (2016): e0161749, https://doi.org/10.1371/journal.pone.0161749; Geert A. Buijze et al., "Correction: The Effect of Cold Showering on Health and Work," PLOS One 13, no. 8 (2018): e0201978, https://doi.org/10.1371/journal.pone.0201978.

“walls.” As mentioned above, pinpointing an exact temperature for every reader isn’t possible because it will vary according to differences in the amount of body fat, time of day, tolerance for cold, and other factors. That being said, adrenaline levels reliably increase when we expose our body surface to temperatures below 59 degrees Fahrenheit (15°C) and the colder the water, the faster and larger the resulting catecholamine response will be. This relationship isn’t linear. We cannot say, “Sit in this temperature water for this long and you’ll get this much more adrenaline and dopamine released.” It doesn’t work like that. We don’t work like that. Nor can we say that x number of minutes in x degrees water will elevate our dopamine x amount for x number of hours after.

That’s why, although I refer to deliberate cold exposure as a protocol, like resistance training or cardio, you will need to consider what number of sets, so to speak, works best for you. It is best thought of as a time to practice triggering specific types of adaptations to build resilience. Some people say you should try to relax and breathe slowly during deliberate cold exposure. Others suggest distracting yourself with mental arithmetic or singing a song. I disagree and view both those approaches as the loss of a valuable opportunity to focus on stress control. You can take this time to learn to steer your mind and body through the stress response.

Resilience, grit, and mental toughness are really about the ability to stay in control of your thoughts, words, and actions when you experience that stress response. When you get into the cold, your prefrontal cortex experiences a surge in norepinephrine input, which causes a disruption in its communication with subcortical areas.[90] This is true of any stressful circumstance, whether deliberate or involuntary. With your prefrontal cortex function essentially impaired for the first thirty seconds after you get in the cold, all your thoughts and actions initially will default to removing the stressor. It's where the immediate "I want out!" response comes from. The key is to remind yourself that it is normal to feel like that. You're in what I call "pure impulse" mode. Do this: Try to consciously watch your mind shift from "I want out!" to "I've got this," which will likely happen again and again very rapidly in that first thirty or so seconds. By observing that phase and not trying to grind your way through it, you are actively engaging your prefrontal cortex in a way that is not typical of someone going through a mild cold-shock response. That's an extremely valuable skill, to be able to observe your own mind during stress. It's not about passively observing or dissociating, nor is it about

[90] Amy F. T. Arnsten, "Stress Signalling Pathways That Impair Prefrontal Cortex Structure and Function," Nature Reviews Neuroscience 10, no. 6 (June 2009): 410–22, https://doi.org/10.1038/nrn2648; Shabnam Hossein et al., "Effects of Acute Stress and Depression on Functional Connectivity Between Prefrontal Cortex and the Amygdala," Molecular Psychiatry 28, no. 11 (April 2023): 4602–12, https://doi.org/10.1038/s41380-023-02056-5.

getting hijacked by stress. It's about learning adaptive self-control during stressful moments.

The first time you try this, it's hard, but most people find that within a few sessions the mind racing starts to lessen, and pretty soon, they feel in control of their thoughts and actions from the start. Few people have mastered this valuable skill in any domain of life, but anyone can learn it. Deliberate cold provides a safe way to practice.

I believe the walls approach mentioned above is the best way to practice this skill, whose benefits translate to other forms of stress. When we measure time elapsed in the cold, we focus, not surprisingly, on its passage. As you learned in the introduction to this chapter, high arousal states tend to slow our perception of that. We count, we wonder why time seems to be moving so slowly—a known function of stress—and we focus on it… 3, 2, 1, okay, done! Real-life stressors are very different. Some hit when we least expect it (e.g., an accident or crisis), others are stressful in large part because we know they are coming (e.g., a public talk we need to give, a medical appointment we would sooner avoid, etc.). The way to train for real-life stress in the cold is therefore to learn to sense the increases in adrenaline that occur as we approach, enter, stay in, and exit it. Each of these can be the walls you need to climb over. If you think like that, you'll notice something very useful: Stress is not

constant. It's not, as we say, a step function. It oscillates, often very quickly and irregularly. The walls appear, and then disappear and reappear with unpredictable frequency.

If you dread the cold, remember that adrenaline is released merely from the anticipation of getting in. Congratulations! That's the first wall. And as we lower ourselves into the cold, that's another wall; next comes the first "I want out!" wall, then it passes… then comes the next wall, the next, and the next. Importantly, the time between walls is variable. This is also important to notice, as that is exactly how real-life stress feels. We often can't predict when the next stressful event is coming.

Basically I'm suggesting you approach deliberate cold exposure as resilience training. And by doing it you'll become mentally stronger in the face of nonvoluntary stress. Just be sure you don't stay in the cold longer than five minutes. Stick to the one to three minutes recommended here, or "go over" two to eight walls. Then get out and go about the rest of your day. Some people, myself included, like to take a hot shower after deliberate cold exposure. People debate whether it's better to end on cold or not. I don't care. I like a hot shower after and, in my experience, it does not seem to diminish any of the benefits. Find what works for you.

As a final point, you actually have more moment-to-moment control over the temperature of the water than you may think. While you can't quickly make the water temperature change, you can dynamically adjust your experience of it. If you hug your arms around your body, raise your shoulders toward your ears and bring your knees up toward your chest, you will feel slightly warmer. Conversely if you keep your arms and legs extended and move them around, you'll break up the thermal layer of heated water around your body, making the experience much colder. Want it even colder? Make sure to let the cold water reach as high as your armpits and keep your hands and feet under. You'll want to remind yourself that this is all very good for you now, and in the hours to come. Cursing Selye, as I do, or cursing me (feel free), is allowed and encouraged.

COMMON QUESTIONS

Is deliberate cold exposure safe

Cold exposure can be dangerous if the water is exceedingly cold. Exercise caution and prudence with this tool. If you are new to cold immersion, start off slowly, with short exposures in temperatures that are outside your comfort zone but not extremely so. Try a cold shower, which allows you to adjust the temperature higher or lower quickly. Alas, there is not much data on deliberate cold exposure for pregnant women so if you're concerned about potential negative outcomes,

avoid this. Don’t do it when sick with a cold, the flu, or other respiratory illnesses. Regardless of what you may see online, it’s likely to make your illness worse. Here’s a good set of guidelines for nonpregnant individuals with no serious health issues:

- Start with water around 50 degrees Fahrenheit the first time, maybe in the mid- to high 40s. Test the water temperature with your hand or foot before getting in; it should feel cold and make you think, “Whoa, that’s cold!” but it should not “burn.”
- If the water triggers a reflex to raise your shoulders and huddle your arms when you get in, then it’s cold enough. If your breathing involuntarily quickens, it’s cold enough.
- Get in up to your neck with your feet under. Put your hands under last.
- If you opt for a cold shower, back into the water until your whole body and head are underneath the stream.
- Expect your mind to start racing with thoughts like, “I want out! Why am I doing this?!” Congratulations, you have a functional autonomic nervous system.

Note: Never, ever jump or dive into very cold water. Lower yourself in while holding something reliably stationary like a ladder or the sides of a cold plunge, or back into a cold shower. I recommend against deliberate cold exposure in cold or frozen outdoor bodies of water unless you have an

expert with you, know the location well, and there are real safety measures in place. There is a video on the internet of a woman who drowned after jumping into an ice hole; unbeknownst to her, there was a rapid current under the ice and she was swept under. Her husband allegedly went in after her and also drowned. I have a professor friend who nearly drowned doing deliberate cold exposure in a river. He had done at-home cold plunges many times before but hadn't realized how cold the river was until he swam out a few feet from shore. He couldn't breathe properly in a shocked state and he was alone. Have someone with you and decide on a nonverbal signal for "Save me!" before you get in.

Never—never, ever!—combine immersion in cold water with any type of deliberate hyperventilation before or during. Your breathing will naturally quicken when you get in cold water, but what I am referring to is deliberate hyperventilation (Wim Hof or Tummo breathing, as it is sometimes called). This breathing method causes the lowering of carbon dioxide and will delay the gasp reflex, potentially leading you to black out and drown; sadly, people have died this way.

Deliberate cold exposure is easy to do safely. Almost all deaths from it occur because people were reckless in some way or combined it with hyperventilation. Don't be reckless.

Are there additional benefits to deliberate cold exposure

A study from metabolic scientist Susanna Søberg, at the University of Copenhagen, and colleagues showed that if people did eleven minutes total per week of deliberate cold exposure, divided into two to four sessions, they enhanced the amount of brown fat in their bodies, a healthy, mitochondria-rich fat that surrounds the heart, neck, and clavicles.[91] That additional brown fat allowed them to feel more comfortable in other cold environments, because when the sympathetic nervous system is active it causes brown fat to burn fatty acids and glucose, which generates heat in the process. In that way, brown fat is like an internal furnace. So if you're someone who is often cold, deliberate cold exposure can help shift your thermal set point. This is connected to the metabolic benefits of cold immersion, which are outside the scope of this protocol. (Note that the participants didn't gain weight; they just added this special—and beneficial—type of fat.)

I hate the cold so much. Is there any alternative method

Yes. As an alternative to deliberate cold exposure, you can do three to five minutes of cyclic hyperventilation away from water, which reliably raises adrenaline levels. Cyclic

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[91] Susanna Søberg et al., "Altered Brown Fat Thermoregulation and Enhanced Cold-Induced Thermogenesis in Young, Healthy, Winter-Swimming Men," Cell Reports Medicine 2, no. 10 (2021): 100408, https://doi.org/10.1016/j.xcrm.2021.100408.

hyperventilation, as the name suggests, involves taking long, vigorous inhales through your nose and passively exhaling through your mouth—think of it as letting the air fall out rather than forcing it out—over and over. It's not easy to keep up this breathing for several minutes. You'll find it makes you quite agitated and a bit lightheaded.

Data suggest that cyclic hyperventilation works similarly to cold exposure.[92] Adrenaline rises. Your heart rate increases. Your limbs will feel like they want to move. And your brain will ask "Why am I doing this?!" It's normal to feel some light burning in your nostrils as well. It feels uncomfortable while you're doing it, but it can improve mood and sleep afterward. Also, you can back off the intensity and/or duration of the inhales and thereby modulate your stress response as you go, an advantage that cold exposure doesn't offer.

So if you really hate the cold or if you don't have access to it, cyclic hyperventilation is the next-best alternative. Just never do it in combination with deliberate cold exposure as, again, you can pass out and drown. People have died combining cyclic hyperventilation with cold immersion;

[92] Balban et al., "Brief Structured Respiration Practices"; Susanna Søberg et al., "Altered Brown Fat Thermoregulation and Enhanced Cold-Induced Thermogenesis in Young, Healthy, Winter-Swimming Men"; Matthijs Kox et al., "Voluntary Activation of the Sympathetic Nervous System and Attenuation of the Innate Immune Response in Humans," Proceedings of the National Academy of Sciences 111, no. 20 (May 2014): 7379–84, https://doi.org/10.1073/pnas.1322174111.

that's why I'm repeating the warning here yet again. And of course never do cyclic hyperventilation while driving or operating machinery, or other activities that demand your attention. Just sit or lie down somewhere safe for a few minutes to use this tool.



STRESS CONTROL PROTOCOL

Do Non-Sleep Deep Rest (NSDR) to Buffer Against Stress

WHAT TO DO

- Do a non-sleep deep rest (NSDR) session for 10 to 30 minutes. Sit or, ideally, lie down in a quiet place and listen to a scripted NSDR session. Guided NSDRs in different voices are available online (hubermanlab.com/nsdr) for zero cost and can be downloaded to use offline (also at no cost). NSDR is especially useful in high-stress times lasting weeks or longer, such as intense work or school periods, or when facing deadlines and life transitions, such as raising young kids or caring for an elder. Ideally you do NSDR once per day but it is beneficial to reduce stress at any point. I do it at least once a week, depending on my schedule.

HOW IT WORKS

I talked about NSDR earlier in this book (remember Sleep Protocol 8?). It is impossible to overstate what an extraordinarily potent tool NSDR can be for mental and physical health. In chapter 1, I covered NSDR's ability to restore mental and physical energy without creating sleep inertia grogginess. But it is particularly potent when it comes to limiting unhealthy stress. I was first introduced to this tool nearly ten years ago. I often get asked "What is something you wish you had known when you were younger?" and my answer is always, "I wish I had known about yoga nidra (aka NSDR)."

NSDR has so many benefits. It replenishes your mental and physical energy,[93] it can offset many of the negative effects of sleep deprivation,[94] it improves the depth and continuity of your sleep at night,[95] and it buffers you against the

[93] Omar Boukhris et al., "The Acute Effects of Nonsleep Deep Rest on Perceptual Responses, Physical, and Cognitive Performance in Physically Active Participants," Applied Psychology: Health and Well-Being 16, no. 4 (2024): 1967–87, https://doi.org/10.1111/aphw.12571.

[94] Vaishnu Karthik Kamaraj et al., "Mechanistic Insights from Neurophysiological Studies of OM Chanting, Pranayama and Yoga Nidra on Sleep Quality," Annals of Neurosciences (October 29, 2025): 09727531251386644, https://doi.org/10.1177/09727531251386644; Erica Sharpe et al., "A Closer Look at Yoga Nidra—Early Randomized Sleep Lab Investigations," Journal of Psychosomatic Research 166 (March 2023): 111169, https://doi.org/10.1016/j.jpsychores.2023.111169.

[95] Abhijit Dutta et al., "Efficacy of Yoga Nidra in Managing Sleep Disorders: A Systematic Review of Randomized Controlled Trials," Journal of Integrative

harmful effects of both acute and chronic stress.[96] To get the most out of this practice, I'll make a few things clear. First, NSDR is not a nap. Nor is it hypnosis. Nor is it meditation, as I explained in chapter 1. Rather, NSDR belongs to a category of techniques that can be broadly thought of as self-directed relaxation. You shift your brain and bodily state from one of "alert and ready to move" to "alert yet deeply relaxed." It emphasizes parasympathetic nervous system (PNS) activity, or increases your parasympathetic tone, tilting you toward that calmer side of the autonomic seesaw that I described in the introduction to this chapter. Sounds like a shallow nap… but it's not. Because you deliberately stay awake during this practice, your sympathetic nervous system remains engaged too. It's a unique brain state that has important implications.

Research shows that during NSDR, parts of your brain enter sleeplike states, while other brain areas remain in a state of

and Complementary Medicine 32, no. 3 (2025), https://doi.org/10.1177/27683605251390728.

[96] Shashank Ghai et al., "Effects of Yoga Nidra on Stress, Anxiety, and Depression: A Systematic Review and Meta-Analysis," Annals of the New York Academy of Sciences 1556, no. 1 (2026): e70149, https://doi.org/10.1111/nyas.70149; Boukhris et al., "Acute Effects of Nonsleep Deep Rest"; Karuna Datta et al., "Improved Sleep, Cognitive Processing and Enhanced Learning and Memory Task Accuracy with Yoga Nidra Practice in Novices," PLOS ONE 18, no. 12 (December 2023): e0294678, https://doi.org/10.1371/journal.pone.0294678.

alertness.[97] We don't quite know what to call the overall brain state that NSDR creates, but we know that it's self-directed and has significant stress-reducing and energy-replenishing effects.[98] We also know that people typically emerge from NSDR in an alert-but-calm state of mind and body, that center of the seesaw. When one considers that it costs nothing but ten to thirty minutes of time, that is quite a return on investment!

As discussed in chapter 1, NSDR involves shifting one's focus throughout the session, which for some people makes it easier than meditations that demand you anchor your focus or "clear your mind." NSDR asks that you shift from thinking, doing, and planning, to being and feeling—I know, that sounds super New Agey. But the instruction is based in physiology, not mysticism. Here is why: Turning off our thoughts is really hard. In fact, we can't do it. Even in sleep, we dream. Even in REM sleep people can solve and respond

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[97] Prakash Chandra Kavi, "Conscious Entry into Sleep: Yoga Nidra and Accessing Subtler States of Consciousness," Progress in Brain Research 280 (2023): 43–60, https://doi.org/10.1016/bs.pbr.2022.12.012; Karuna Datta et al., "Electrophysiological Evidence of Local Sleep During Yoga Nidra Practice," Frontiers in Neurology 13 (July 2022): 910794, https://doi.org/10.3389/fneur.2022.910794.

[98] Esther N. Moszeik et al., "The Effects of an Online Yoga Nidra Meditation on Subjective Well-Being and Diurnal Salivary Cortisol: A Randomised Controlled Trial," Stress & Health 41, no. 3 (2025): e70049, https://doi.org/10.1002/smi.70049.

to math problems.[99] (If you're wondering how the experimenters knew that, given people are paralyzed during REM, the subjects signaled correct answers to simple subtraction problems using eye movements.) So forget trying to destress by not thinking. It just won't happen.

We can, however, easily shift our immediate experience from planning and analyzing to just experiencing physical sensations. NSDR has us do that. We deliberately shift our attention from one aspect of our sensations in that moment—for instance from our toes to our fingers, then to a location away from our body such as to sounds in the room, then to sounds far away, then back to a specific location on our body, and so forth. This regular and deliberate shifting of our focus on different aspects of our perception both eliminates the strain of trying to continually focus on one thing, and the tendency of our mind to default to planning and analyzing. With NSDR, your mind drifts from one sensation to the next in an intentional but relaxed way. When you do that, your mind goes into an atypical, almost liminal state, the one described above—somewhere between that of deep sleep and being awake. Cortisol levels drop, discrete brain areas fall asleep while others remain active.[100]

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[99] Karen R. Konkoly et al., "Real-time Dialogue Between Experimenters and Dreamers During REM Sleep," Current Biology 31, no. 7 (April 2021): 1417–27, https://doi.org/10.1016/j.cub.2021.01.026.

[100] Datta et al., "Electrophysiological Evidence of Local Sleep"; Moszeik et al., "Effects of an Online Yoga Nidra Meditation."

Because NSDR is done lying down with eyes closed, from the outside it appears you are napping but you're not. You're in self-directed reset mode.

If you've never done yoga nidra or NSDR before, this may all sound a bit abstract, but it's really very concrete. It is grounded in what we know about self-directed brain states. And it's very easy to do. One of the biggest advantages of NSDR is that, in contrast to mindfulness meditation—which is a skill you build up gradually over time and that requires daily (or near daily) practice to maintain the positive effects—NSDR can work the first time, and every time.

You lie down, listen to the instructions, and voila! Ten to thirty minutes later you emerge feeling renewed and rested. Most everyone who tries it marvels at the immediate positive effects. It's starting to be used in schools, by elite military operators, professional athletes, Fortune 500 CEOs, and many other men and women around the world as a tool for managing stress and even as a means to accelerate recovery from illness and injury, to offset some of the negative symptoms of menopause, sleep deprivation, trauma, and more.[101] I encourage everyone to try it a few times and experience the benefits for themselves.

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[101] Ghalia Akram, "Testing Non-Sleep Deep Rest (NSDR) Protocol to Counter the Cognitive Consequences of Short Sleep in Young Adult Students" (master's thesis, Harvard University Division of Continuing Education, 2024), https://nrs.harvard.edu/URN-3:HUL.INSTREPOS:37378464; Jamie Mobley, "iRest, You Rest: Virtual Yoga Nidra Sessions to Help Fight Stress," VA News,

COMMON QUESTIONS

When should I do NSDR

This is the most common question I get about NSDR. You can do it ANY time. In the morning if you wake up and don't feel fully rested and don't have time to get another hour or more of sleep. In the afternoon instead of a nap and if you know you'll be working harder or longer than usual that day. Before a workout if you're feeling less-than-motivated or lacking energy. In the evening if you get home and want to destress from the day. In the middle of the night if you wake up and can't fall back asleep. (In that case, get out of bed and do an NSDR then get back into bed; a return to sleep will arrive easier.)

October 23, 2019, https://news.va.gov/67404/irest-rest-virtual-yoga-nidra-sessions-help-fight-stress/; Robert Mays, "The NFL's Mindfulness Movement Is Spreading," Ringer, August 15, 2017, https://www.theringer.com/2017/08/15/nfl/mindfulness-training-falcons-colts-49ers-dan-quinn; Lane Florsheim, "Google CEO Sundar Pichai's Vision for Return to Work," Wall Street Journal, February 28, 2022, https://www.wsj.com/lifestyle/careers/google-ceo-sundar-pichais-vision-for-return-to-work-11646055640; Adam Wrzeciono et al., "Non-Sleep Deep Rest Relaxation and Virtual Reality Therapy for Psychological Outcomes in Patients with Coronary Artery Disease: A Pilot Randomized Controlled Trial," Journal of Clinical Medicine 13, no. 23 (2024): 7178, https://doi.org/10.3390/jcm13237178; Kamalesh K. Gulia and Sapna Erat Sreedharan, "Yogic Sleep and Walking Protocol Induced Improvement in Sleep and Wellbeing in Post-Menopausal Subject: A Longitudinal Case Study During COVID Lockdown," Sleep Vigil 6, no. 1 (2022): 229–33, https://doi.org/10.1007/s41782-021-00180-2; L. Stankovic, "Transforming Trauma: A Qualitative Feasibility Study of Integrative Restoration (iRest) Yoga Nidra on Combat-Related Post-Traumatic Stress Disorder," International Journal of Yoga Therapy 21 (2011): 23–37, https://doi.org/10.17761/ijyt.21.1.v823454h5v57n160.

The simple answer I default to whenever asked “When should I do NSDR?” is “First thing in the morning, or in the afternoon, or any time of night…” It is that versatile and that useful.



STRESS CONTROL PROTOCOL

Do a Micro Meditation Immediately After Waking Up

WHAT TO DO

- Incorporate a simple, five-minute daily meditation into your routine. Any time of day can be beneficial, but doing it before getting out of bed each morning ensures you don’t have to carve out time later and may offer distinct benefits as well.
- If you feel an urgency to get out of bed before five minutes have elapsed, stay put. Contrary to how it is usually framed, meditation is one of the best forms of resilience training. So see this as an opportunity to practice resilience. This meditation is not about clearing the mind. It’s about building the capacity to sit with the stress that even micro meditation can trigger. This will help you better deal with real-life stressors once your day starts.

HOW IT WORKS

Plenty of evidence exists that mindfulness-based meditation practices buffer us against stress, and even brief daily sessions reduce anxiety.[102] I mention both stress and anxiety here, because there's a difference. Some say stress has an identifiable cause, or a known stressor, whereas anxiety does not. That's a good working distinction. But the fact that meditation protects us against both highlights that each experience is based in the same thing: an elevated sympathetic tone that exceeds the demands of the stressor.

Meditation can adjust your autonomic seesaw, bringing you back to the center. But just how may surprise you.

Most everyone assumes the goal of a meditation practice is to quiet the mind and that it involves cultivating a sense of peace during the practice. In some sense that's true. One study found that four consecutive days of a meditation practice reduced anxiety and increased mindfulness, a broad term encompassing a sense of awareness, calm, and being in the present moment.[103] A 2019 report by New York University neuroscientist Wendy Suzuki and colleagues

[102] Paul Grossman et al., "Mindfulness-Based Stress Reduction and Health Benefits: A Meta-Analysis," Journal of Psychosomatic Research 57, no. 1 (2004): 35–43, https://doi.org/10.1016/S0022-3999(03)00573-7; Balban et al., "Brief Structured Respiration Practices."

[103] Fadel Zeidan et al., "Mindfulness Meditation Improves Cognition: Evidence of Brief Mental Training," Consciousness and Cognition 19, no. 2 (2010): 597–605, https://doi.org/10.1016/j.concog.2010.03.014.

found that doing eight weeks of thirteen-minute daily meditation notably decreased anxiety.[104]

But data also show that meditation can make us more resilient by allowing us to observe stress when it hits without reacting to it. When psychologist Richard Davidson—one of the world's foremost experts on the science of meditation—appeared on my podcast, he described that his lab saw a statistically significant increase (!) in people's anxiety levels in the first week that they started doing a regular meditation practice.[105] Wait, what? Meditation increases anxiety? Yes. Because it can be maddening to force yourself to focus and refocus on your breathing over and over for five minutes when you wake up and you have to dive into your day. Your mind wants to shift to planning and doing. In that way, meditation can serve as a way to increase your stress threshold. Not unlike learning to stay in uncomfortably cold water.

This is as hard as it sounds, but well worth it. Regular meditation teaches us to observe, rather than act on, our autonomic arousal. It also reduces the likelihood that real-life stressors will hijack our mind. Just like Stress Control Protocol 2.

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[104] Basso et al., "Brief, Daily Meditation Enhances Attention."

[105] Andrew Huberman, "Science-Based Meditation Tools to Improve Your Brain and Health | Richard Davidson," Huberman Lab (podcast audio), March 6, 2023, https://www.hubermanlab.com/episode/science-based-meditation-tools-to-improve-your-brain-and-health-richard-davidson.

One study found that the intensity of the emotions people experienced while viewing potentially disturbing pictures was reduced when people did so from a “mindful” state, indicating that meditation promotes a kind of awareness that makes a person’s autonomic nervous system less subject to outside stimuli.[106] Indeed, neuroimaging studies have confirmed that meditation consistently alters brain regions connected to both body awareness and mental awareness, as well as emotional regulation.[107] The underlying mechanism appears to be connected to the production of GABA, an inhibitory neurotransmitter—meaning a chemical that generally suppresses the firing of other neurons. People with panic disorders and anxiety tend to have decreased GABA activity in particular regions of the brain.[108] Researchers have theorized that the increased blood flow to the prefrontal cortex that seems to occur during certain types of meditation may, in turn, lead to increased GABA activity, along with improved top-down inhibition of brain areas that generate

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[106] Véronique A. Taylor et al., “Impact of Mindfulness on the Neural Responses to Emotional Pictures in Experienced and Beginner Meditators,” NeuroImage 57, no. 4 (2011): 1524–33, https://doi.org/10.1016/j.neuroimage.2011.06.001.

[107] Kieran C. R. Fox et al., “Is Meditation Associated with Altered Brain Structure? A Systematic Review and Meta-Analysis of Morphometric Neuroimaging in Meditation Practitioners,” Neuroscience & Biobehavioral Reviews 43 (June 2014): 48–73, https://doi.org/10.1016/j.neubiorev.2014.03.016.

[108] J. Douglas Bremner et al., “SPECT [I-123]iomazenil Measurement of the Benzodiazepine Receptor in Panic Disorder,” Biological Psychiatry 47, no. 2 (2000): 96–106, https://doi.org/10.1016/s0006-3223(99)00188-2.

reflexive thoughts and actions, and this equates to a general sense of relaxation.[109]

A bit more on the format of the micro meditation: stay in bed or, if you must, get up and use the bathroom first, and then set a five-minute timer, sit or lie down, and stay as still as you can while you focus on your breathing. Observe where your mind goes. If it wanders—which is completely normal by the way—redirect your focus back to your breath. Slow nasal-only inhales and exhales are best. But the most important thing is to do it, and on days when you feel stressed, do it anyway. In fact, that's a bonus opportunity. The benefits to your ability to observe your stress rather than have it engulf you that come from a daily five-minute meditation are well worth it. And of course, if you want to do a longer meditation, go for it.

As simple as this protocol may sound, I and countless others can attest to the fact that it provides immediate benefits, and benefits that extend throughout the day. And one more thing: We don't yet understand the science of exactly how, but it also seems to create the ability to "see" when stress is coming your way. Perhaps it sensitizes the brain circuits that respond to autonomic arousal. Perhaps it leads to better

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[109] Divya Krishnakumar et al., "Meditation and Yoga can Modulate Brain Mechanisms that Affect Behavior and Anxiety—A Modern Scientific Perspective," Ancient Science 2, no. 1 (April 2015): 13–19, https://doi.org/10.14259/as.v2i1.171.

recognition of adrenaline in our body rising, or our heart rate starting to increase. After all, a five-minute meditation will gradually have you feeling more anxious as you progress from start to finish; it's very different from getting into cold water, which causes a big spike in adrenaline right at the start. But that ability to sense approaching stress is a game changer, as it can reduce your reactivity and improve your ability to make decisions about what to do and say—and in many cases, what not to do and say—when the inevitable stressors of life appear.



STRESS CONTROL PROTOCOL

Boost Your Morning Cortisol and Catecholamine Peak

WHAT TO DO

- Stack three to four of the pro-cortisol and pro-catecholamine triggers below in the morning, and limit them at night. Get bright light, ideally sunlight, in your eyes for about ten minutes, as soon as possible after waking. If you wake before the sun is up, you'll either have to wait for the sun to rise or use a 10,000 lux light for this trigger. (I cover how in detail in Sleep Protocol 1.)
- Hydrate with 16 to 32 ounces of water and consume some caffeine in the first two hours of your day. How much

caffeine? 100–400 milligrams depending on your body weight and caffeine tolerance. A strong cup of coffee or yerba maté (my preferred caffeine source) has 150–200 milligrams of caffeine. Note: Chronic caffeine drinkers don't get as much of a cortisol increase from their usual amount of caffeine as those who aren't, but it still has a positive effect on other catecholamines for reasons covered in chapter 1.[110]

- Exercise in the morning. If your schedule allows, do cardiovascular or resistance training within the first two to three hours of waking; the closer to waking the better. (The foundational fitness program in chapter 2 offers a sustainable weekly structure, but of course there are many others out there.) If you can't exercise in the morning, take a brief walk, do some jumping jacks, skip rope, or do some light calisthenics, even for just five to ten minutes. While not a full-fledged workout, it will help stimulate the morning cortisol and catecholamine increase you're after.
- Do one of the deliberate cold exposure protocols discussed in this chapter. Remember, this takes only one

[110] Anna Gavrieli et al., "Caffeinated Coffee Does Not Acutely Affect Energy Intake, Appetite, or Inflammation but Prevents Serum Cortisol Concentrations from Falling in Healthy Men," Journal of Nutrition 141, no. 4 (April 2011): 703–7, https://doi.org/10.3945/jn.110.137323; William R. Lovallo et al., "Caffeine Stimulation of Cortisol Secretion Across the Waking Hours in Relation to Caffeine Intake Levels," Psychosomatic Medicine 67, no. 5 (2005): 734–39, https://doi.org/10.1097/01.psy.0000181270.20036.06.

to three minutes, and a cold shower works too. Just do it before resistance training, ideally not after (refer to Stress Control Protocol 2). As an ALTERNATIVE to deliberate cold exposure, try cyclic hyperventilation: Do a cycle of twenty-five to thirty deep, vigorous inhales through your nose, with each inhale followed by a passive exhale—meaning just let the air fall out of your lungs via your mouth. Afterward, you'll feel a bit lightheaded. That is a normal response. Exhale all your air and hold your breath for thirty to sixty seconds. You should no longer feel lightheaded. You can repeat this process for two to three more rounds. Note: NEVER do cyclic hyperventilation before or during deliberate cold exposure, anywhere near water, while driving, operating machinery, or performing other activities that demand your attention. You may recall from those protocols that deliberate cold exposure doesn't spike cortisol all that much, but it does significantly increase the other catecholamines. And that is beneficial here.

- Consider occasional (once or twice per week) use of low-dose licorice root containing glycyrrhizin or consume grapefruit in the first one to three hours of your day. Both have been shown to significantly increase cortisol and the duration of cortisol action. Subjectively, they can increase feelings of morning and daytime energy and focus. Warning: Despite licorice and grapefruit

sounding natural (and they are), their effects can be very potent. They can increase blood pressure and/or fluid retention at high doses, so use the minimal effective dose and, as with all supplements discussed in this book, consult your doctor first.

HOW IT WORKS

Cortisol and the catecholamines adrenaline and norepinephrine, but especially cortisol, are often thought of purely as stress chemicals. The third catecholamine—dopamine—escaped this reputation but is usually thought of (also incorrectly) as purely a pleasure chemical. And while cortisol is involved in the stress response, thinking about it as purely a stress chemical is wrong. I wish I could banner this across the sky: Cortisol Is Not a Stress Hormone!

In fact, if I could impart just one message about how to improve your health, it would be this: Strive to get your morning cortisol high (yes, high!) and your nighttime cortisol low.

The main function of cortisol is to ensure there is enough energy in the form of glucose, or blood sugar, for you to meet life's demands. Yes, that can include overcoming stressors, but it also includes simply waking up each morning with the physical and mental energy required for

work, school, family, and exercise. Cortisol causes glucose (as well as fats) to get released into the bloodstream, and it also stimulates the liver to make more glucose.[111]

Cortisol and adrenaline play additional key roles in our body's everyday functioning. They improve our immune system's ability to ward off infections, they elevate our mood, they allow for our capacity to form new memories, and they increase metabolism… so just forget the idea that cortisol and adrenaline are only about stress.

A gradual rise in your cortisol levels starts three to four hours before you wake up each morning and its levels eventually hit a threshold level, at which point you wake up, open your eyes, and get out of bed. Keep in mind that cortisol doesn't act alone to rouse you from sleep; a parallel rise in body temperature and clearance of the sleepiness-inducing molecule adenosine follows a similar schedule. Nonetheless, scientists call this rise the "cortisol awakening response" or CAR. Now that makes it sound like a medical problem of some sort, but CAR is a normal and (provided it is timed correctly) health-promoting aspect of our biology.[112] And it helps buffer you against unhealthy stress.

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[111] Jasleen Kaur et al., "Physiology, Cortisol," StatPearls, last updated December 1, 2025, https://www.ncbi.nlm.nih.gov/books/NBK538239/.

[112] Clara Velazquez Sanchez and Jeffrey W. Dalley, "The Cortisol Awakening Response: Fact or Fiction?," Brain and Neuroscience Advances 9 (April 24, 2025), https://doi.org/10.1177/23982128251327712; Tobias Stalder et al., "The

Cortisol first starts to rise as a result of a signal from your SCN, your brain's master circadian clock that we discussed in chapter 1. For most people, that rise starts somewhere around five hours into their night's sleep and continues sharply over the subsequent two to three hours until it reaches a threshold level and bam! You wake up. Even if you use an alarm clock, you can be sure your cortisol levels were rising as you approached the start of your day. Now, once you're awake, cortisol continues to increase until it hits its peak for that twenty-four-hour cycle. For most people, that peak occurs anywhere from 0.5 to 1.5 hours after waking up.

Then cortisol levels slowly start to decline—a process that continues into the late afternoon and evening. Assuming you encounter no major stressors, your cortisol levels continue to drop in the hours before you fall asleep and reach their lowest level about three to four hours after you doze off, and then slowly start to rise again to eventually trigger the next morning's CAR, and so on. This is the ideal cortisol pattern

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Cortisol Awakening Response: Regulation and Functional Significance," Endocrine Reviews 46, no. 1 (2025): 43–59, https://doi.org/10.1210/endrev/bnae024; Angela Clow et al., "The Cortisol Awakening Response in Context," International Review of Neurobiology 93 (2010): 153–75, https://doi.org/10.1016/S0074-7742(10)93007-9; Nicole P. Bowles et al., "The Circadian System Modulates the Cortisol Awakening Response in Humans," Frontiers in Neuroscience 16 (November 2022): 995452, https://doi.org/10.3389/fnins.2022.995452.

for mental and physical health, energy, and resilience to stress.

Night shift workers deal with a very disrupted sleep-wake schedule, and their cortisol pattern gets fragmented, with multiple peaks and troughs throughout the day. For some, the pattern inverts to match their wake-sleep schedule. Although that might seem like not that big of a deal if they are still getting enough sleep, it's known to cause serious and long-lasting health issues. Indeed, many of the known health detriments of nocturnal schedules are due primarily to alterations in the pattern of cortisol release and its downstream effects. I cover what shift workers can do to partially offset those negative effects in chapter 1 and chapter 5.[113]

Most people, even night owls or those staying up too late on their phones, are not completely inverting their sleep-wake schedule, and yet there is reason to believe that they have less than ideal cortisol patterns. I say "most people" because the vast majority of questions I get are about wanting better sleep, more energy, improved focus, and less stress and anxiety. And when I inquire about the structure of their days,

[113] Aikaterini Andreadi et al., "Modified Cortisol Circadian Rhythm: The Hidden Toll of Night-Shift Work," International Journal of Molecular Sciences 26, no. 5 (February 2025): 2090, https://doi.org/10.3390/ijms26052090; Fadime Ulupinar et al., "The Effect of Single and Double Shift Works on Salivary Cortisol Levels in Nurses," Nursing Open 13, no. 4 (April 2026): e70495, https://doi.org/10.1002/nop2.70495.

I learn that very few people are doing the few simple things to ensure high cortisol in the morning and low cortisol at night. This sets them up for health problems and lowered resilience to stress.

Numerous studies show that if your cortisol levels fall too slowly as you approach the later hours of your day—a so-called "flattening of the diurnal cortisol rhythm"—or if your cortisol is spiked by stress or bright light in the evening, you can experience that as increased anxiety, and sleep can suffer too.[114] If this happens once in a while, it's no big deal. Acute stressors—an argument, a small crisis at work, and so on—are part of life. They spike our cortisol and catecholamines. But these levels ordinarily return to baseline a few minutes or hours later. So don't freak out about a bit of stress here or there even late in the day! The stress management tools discussed elsewhere in this chapter will assist in helping you navigate those events.

If, however, your cortisol undergoes stress-induced peaks in the late evening or nighttime, or if your morning cortisol is too low or declines too slowly, that flattening of the cortisol curve is what puts you into a state of chronic stress.[115] This can cause a cascade of detrimental events in your body and

[114] Andreadi et al., "Modified Cortisol Circadian Rhythm."

[115] Ethan S. Young et al., "The Dual Impact of Early and Concurrent Life Stress on Adults' Diurnal Cortisol Patterns: A Prospective Study," Psychological Science 30, no. 5 (2019): 739–47, https://doi.org/10.1177/0956797619833664.

brain. Diabetes, hypertension, depression, and weight gain can result, as can physical imbalance, leaving a person feeling unsteady on their feet.[116] Studies have also linked abnormally high cortisol levels later in the day and into the night to acne, irregular menstrual cycles, and digestion issues.[117] High cortisol at night has been linked to insomnia

[116] Vittoria Favero et al., "The Degree of Cortisol Secretion Is Associated with Diabetes Mellitus and Hypertension in Patients with Nonfunctioning Adrenal Tumors," Cardiovascular Diabetology 22, no. 1 (2023): 102, https://doi.org/10.1186/s12933-023-01836-1; Vittoria Favero et al., "Correction: The Degree of Cortisol Secretion," Cardiovascular Diabetology 22, no. 1 (2023): 127, https://doi.org/10.1186/s12933-023-01863-y; Mahmut Cay et al., "Effect of Increase in Cortisol Level Due to Stress in Healthy Young Individuals on Dynamic and Static Balance Scores," Northern Clinics of Istanbul 5, no. 4 (2018): 295–301, https://doi.org/10.14744/nci.2017.4210; Eline S. van der Valk et al., "Stress and Obesity: Are There More Susceptible Individuals?," Current Obesity Reports 7, no. 2 (2018): 193–203, https://doi.org/10.1007/s13679-018-0306-y; B. R. Walker, "Cortisol—Cause and Cure for Metabolic Syndrome?," Diabetic Medicine 23, no. 12 (2006): 1281–88, https://doi.org/10.1111/j.1464-5491.2006.01998.x; Joshua J. Joseph and Sherita H. Golden, "Cortisol Dysregulation: The Bidirectional Link Between Stress, Depression, and Type 2 Diabetes Mellitus," Annals of the New York Academy of Sciences 1391, no. 1 (October 2016): 20–34, https://doi.org/10.1111/nyas.13217.

[117] Dominika Borzyszkowska et al., "Evaluation of Hormonal Factors in Acne Vulgaris and the Course of Acne Vulgaris Treatment with Contraceptive-Based Therapies in Young Adult Women," Cells 11, no. 24 (2022): 4078, https://doi.org/10.3390/cells11244078; Marzieh Saei Ghare Naz et al., "The Menstrual Disturbances in Endocrine Disorders: A Narrative Review," International Journal of Endocrinology and Metabolism 18, no. 4 (2020): e106694, https://doi.org/10.5812/ijem.106694; Christine E. Cherpak, "Mindful Eating: A Review of How the Stress-Digestion-Mindfulness Triad May Modulate and Improve Gastrointestinal and Digestive Function," Integrative Medicine 18, no. 4 (2019): 48–53, https://pubmed.ncbi.nlm.nih.gov/32549835/.

and reduced amounts of REM sleep.[118] Given this, everyone would do well to try to anchor and amplify their cortisol and catecholamine peak in the first hours after waking to start their day and limit their cortisol and catecholamine levels at night. That simple framework goes a long way toward limiting your chronic stress. And that's what this protocol is all about.

Let's start with what to do at night, because how you prepare for bed affects your cortisol and catecholamine levels the next day. Some of what you're about to read will be familiar to you from chapter 1. Before bed, implement a wind-down routine of dimming the lights, maybe taking a hot shower or bath or even washing your face with hot water. If you're feeling stressed from the day, take a few moments to do some physiological sighs to lower your heart rate, and/or do a ten- to thirty-minute NSDR session. Don't drink too many fluids or eat in the two to three hours before sleep, and definitely try to avoid caffeine in the eight to ten hours prior to bedtime. Viewing bright overhead LED lights (of the sort that are in many bathrooms) is a quick way to spike your cortisol and quash your melatonin, and checking your email or the news in the last hour before sleep is definitely a risky move if you want to keep your cortisol low at night (see

[118] Matias M. Pulopulos et al., "Relationship Between Cortisol Changes During the Night and Subjective and Objective Sleep Quality in Healthy Older People," International Journal of Environmental Research and Public Health 17, no. 4 (2020): 1264, https://doi.org/10.3390/ijerph17041264.

chapter 1 for details). And to be clear: Low cortisol at night should be your goal!

If you happen to wake up in the middle of the night for whatever reason, avoid viewing bright lights or looking at your phone. Again, do some physiological sighs to help keep your heart rate low, and let yourself drift back to sleep. If you can't fall back asleep, listen to an NSDR session. Or try the weird eye-movement trick I shared in chapter 1.

By the way, if you exercise in the evening or at night, that does significantly increase late-day cortisol. It's better to exercise than not to, if you have a workout planned. To help keep your cortisol and catecholamine rhythms on a diurnal schedule, you can reduce the intensity of your evening training a bit and/or be sure to include a wind-down routine afterward. Andy Galpin, an exercise physiologist, encourages this after all workouts, but especially nighttime ones. Even moderate-intensity exercise can cause a large increase in cortisol (as discussed in chapter 2), especially if you consume caffeine first.[119] That is one more reason to drink caffeine before moderate to intense exercise in the morning, but also to avoid it before moderate to intense exercise at night.

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[119] C. Martin Beaven et al., "Dose Effect of Caffeine on Testosterone and Cortisol Responses to Resistance Exercise," International Journal of Sports Nutrition and Exercise Metabolism 18, no. 2 (April 2008): 131–41, https://doi.org/10.1123/ijsnem.18.2.131.

I don't want to demonize late-day exercise. Better to do it than not do it. But just know that if you exercise intensely within four to six hours of bedtime, your next-day waking cortisol may be lower due to the negative feedback regulation of cortisol.[120] That's a mechanism you often see with hormones in the body, like cortisol or testosterone: When those hormones reach a certain high level, it signals specific brain areas and glands, such as the hypothalamus and pituitary and adrenals, to reduce their release of those hormones. Just another reason why spiking your morning cortisol is a good thing. It not only sets you up for the day, it helps keep cortisol levels low at night.[121] So if you exercise at night and wonder why you're dragging a bit the next morning, this may very well be why. Be diligent about using the tools in this protocol and you'll be A-OK.

What about spiking your cortisol and catecholamines in the morning—how best to do that? There too you have options: Viewing bright light can boost your morning cortisol peak by 50 percent, making it a very easy, effective and beneficial practice (see Sleep Protocol 1 for more). Sunlight is ideal,

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120 Travis Anderson et al., "The Effect of Acute Exercise on the Cortisol Awakening Response," European Journal of Applied Physiology 123 (January 2023): 1027–39, https://doi.org/10.1007/s00421-023-05132-4.

121 Peter Y. Liu, "Rhythms in Cortisol Mediate Sleep and Circadian Impacts on Health," Sleep 47, no. 9 (September 2024): zsae151, https://doi.org/10.1093/sleep/zsae151.

but a 10,000 lux light works too.[122] Hydration, caffeine, moderate to intense exercise, deliberate cold exposure, maybe cyclic hyperventilation (but never with deliberate cold exposure): Doing one of these is better than none, two is better than one, and so on. If I had to place them in order of effectiveness, I would say: sunlight viewing, exercise, and caffeine are the big levers here, but don't worry about the order. Let logic be your guide. Obviously hydrate before and during your exercise, drink caffeine for an energy boost, get the sunlight before or during exercise if you can, but if the sun isn't up and you're ready to train, train! Don't wait for the sun to rise, thinking that it has to come first because it's a big lever. The cortisol and catecholamine systems are primed for release in the early hours of your day and will synergize without the need to get the relative timing (of all the things that cause their release) perfect.

When people ask me, "Who has time for all this?" I respond that it takes mere seconds to drink 16 ounces of water, a few minutes to make a caffeinated beverage and step outside into the sun while you drink it, and then depending on your schedule, exercise. Do not obsess over the details of relative timing of these tools.

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[122] Petrowski et al., "Effects of Post-Awakening Light"; Brown et al., "Recommendations for Daytime, Evening, and Nighttime Indoor Light."

A number of people have asked me over the years: “Given the importance of ensuring elevated morning cortisol, is there anything I can take to boost it besides caffeine?”

In chapter 6, I cover the many prescription drugs and supplements that can increase dopamine and norepinephrine, but when it comes to things that boost cortisol there are two that are particularly useful. And they are inexpensive. And commonplace. You may even have one of them in your kitchen right now.

They are grapefruit and licorice root extract. Not licorice candy, but licorice plant root extract.[123]

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[123] Marjolein Drent et al., “Grapefruit Juice Facilitates Cortisol Replacement Therapy: Role of CYP3A Variant Alleles,” Archives of Clinical and Medical Case Reports 5, no. 5 (2021): 656–64; Ya-Jie Xiao et al., “Effects of Grapefruit Juice on Cortisol Metabolism in Healthy Male Chinese Subjects,” Food and Chemical Toxicology 74 (December 2014): 85–90, https://doi.org/10.1016/j.fct.2014.09.012; Christopher Kargl et al., “11β-hydroxysteroid Dehydrogenase Type-II Activity Is Affected by Grapefruit Juice and Intense Muscular Work,” Archives of Endocrinology and Metabolism 61, no. 6 (2017): 556–61, https://doi.org/10.1590/2359-3997000000296; Y. S. Lee et al., “Grapefruit Juice and Its Flavonoids Inhibit 11 Beta-Hydroxysteroid Dehydrogenase,” Clinical Pharmacology and Therapeutics 59 (1996): 62–71, https://doi.org/10.1016/S0009-9236(96)90025-9; Y. Kageyama et al., “Glycyrrhizin Induces Mineralocorticoid Activity Through Alterations in Cortisol Metabolism in the Human Kidney,” Journal of Endocrinology 135, no. 1 (1992): 147–52, https://doi.org/10.1677/joe.0.1350147; M. T. Epstein et al., “Licorice Raises Urinary Cortisol in Man,” Journal of Clinical Endocrinology and Metabolism 47, no. 2 (August 1978): 397–400, https://doi.org/10.1210/jcem-47-2-397; Yu-Jin Kwon, “A Review of the Pharmacological Efficacy and Safety of Licorice Root from Corroborative Clinical Trial Findings,” Journal of Medicinal Food 23, no. 1 (January 2020): 12–20, https://doi.org/10.1089/jmf.2019.445; Paal Methlie et al., “Grapefruit Juice and Licorice Increase Cortisol Availability in Patients with Addison’s

Each can increase the size and duration of your morning cortisol peak and thereby increase your level of alertness and the other subjective effects associated with elevated sympathetic tone. How is this possible? Well…

Grapefruit impacts cortisol levels by two known mechanisms. First, it inhibits the enzyme (11β-HSD2) that blocks cortisol to cortisone conversion—cortisol being the chemical that imparts the functions we've been talking about so far in this protocol. Slower conversion means more active cortisol stays in our system for longer to impart its effects. Second, grapefruit inhibits the enzyme CYP3A4, which blocks the liver's metabolism of cortisol, the net effect of which is to increase cortisol's duration of action.

Licorice root extract contains something called glycyrrhizin (pronounced gly-cer-ry-zin) that potently inhibits the same abovementioned enzyme 11β-HSD2.

You may be thinking, Grapefruit and licorice?! Is he serious?! Yes, quite serious. There are human data to support the use of each to increase cortisol's duration of action including studies demonstrating dose-dependent effects. So much so that well-informed MDs will warn patients with Cushing's syndrome—which is a condition marked by pathologically elevated levels of cortisol—to avoid

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Disease," European Journal of Endocrinology 165, no. 5 (November 2011): 761–69, https://doi.org/10.1530/EJE-11-0518.

grapefruit, licorice root extracts, teas containing glycyrrhizin, and any licorice candy. (Note: Some traditional black licorice root candy, typically sold in Scandinavian countries but elsewhere too, contains glycyrrhizin, though versions without glycyrrhizin exist too.) I suggest most everyone avoid licorice root extract consumption late in the day and at night. If you want to experiment with taking licorice root extract that contains glycyrrhizin please talk to your doctor first. At high doses it can raise blood pressure, cause fluid retention, and even affect the metabolism of certain pharmaceutical drugs and potassium because of the many roles of the abovementioned enzymes in the liver.

I use licorice root extract with glycyrrhizin as a tool occasionally (once or twice a month). I find that 100 to 250 milligrams of licorice root in capsule form is enough to feel its effects. I make sure to use a version containing glycyrrhizin, and I take it with my morning caffeine usually before a hard workout that I need to start earlier than is typical for me. The energy-increasing effect is very noticeable. On very rare occasions I will take it around noon to try to extend my window of elevated cortisol to work for an especially long day, or to exercise hard midday. Again, I use it as a sometimes tool, not a regular tool, to boost cortisol in the morning.

Overall, grapefruit and grapefruit juice have a milder, but still meaningful, impact on cortisol's duration of action. Most people can consume these without an adverse effect. But they can interact with various medications, so be sure to consult your doctor before taking. If you'd like to try incorporating grapefruit as a cortisol-boosting tool in your routine, perhaps eat a grapefruit or drink a small glass of grapefruit juice in the morning (pink or conventional is fine; it doesn't matter). See how that affects your energy throughout the day. I would avoid these at night for obvious reasons.

One more thing to note here is that it is hard to find very-low-dose licorice root extract with glycyrrhizin. Most are 400 milligrams or more, and even at 200 milligrams it works very well to increase energy, so I empty half a 400 milligrams capsule into water or my coffee, and drink that. It tastes good. Remember: It raises blood pressure, and taking glycyrrhizin every day can lead to serious problems over time, so if you experiment with it, try a low dose—even if it means having to open up a capsule to remove a portion of the contents. Once more, talk to your doctor before you do. And if you have high blood pressure, don't take it!

I can't end this protocol without one more reminder about caffeine: If you consume caffeine in the late afternoon, too, don't overdo it. I get it; I love yerba maté in the morning,

and I love, love, love the taste of coffee in the afternoon. But if you're an afternoon caffeine drinker and having sleep issues or trouble getting going in the mornings, you can bet it's at least partially because the caffeine levels in your bloodstream are still too high from your afternoon coffee or tea. Try to drink only decaf after 2:00 p.m., or limit yourself to roughly 60 milligrams of caffeine (that's about 3 to 4 ounces of regular coffee, or an espresso shot), and do your best to avoid it altogether within eight hours of bed (refer to Sleep Protocol 11 for more guidance). You could also opt to ditch the afternoon caffeine and instead to take 300 to 900 milligrams of alpha-GPC, which is a short-acting stimulant that can increase energy and focus and may actually improve REM sleep at night if taken six to ten hours before bedtime. I discuss this tool more in chapter 6.

This protocol includes several tools, so to recap: By increasing the slope and height of your morning cortisol and catecholamine peak, you buffer yourself against later-day stress in several ways. First, you're likely to experience improved energy and focus for many hours afterward.

Viewing bright light,[124] consuming caffeine,[125] exercise,[126] and cold exposure[127] all increase cortisol and/or catecholamine release. This helps us keep our reactions proportional to any stressors we may encounter later. By starting your day in a state of focus and readiness, if a stressful situation arises you don't need to produce more cortisol to confront it effectively. Researchers who have examined the link between morning cortisol and stress call this connection the CAR anticipation hypothesis.[128] It suggests that a big cortisol peak in the morning isn't just to wake us up; it also prepares us for whatever stressors lie ahead.[129]

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[124] Isabella Robertson-Dixon et al., "The Influence of Light Wavelength on Human HPA Axis Rhythms: A Systematic Review," Life 13, no. 10 (2023): 1968, https://doi.org/10.3390/life13101968.

[125] Elli L. Cole et al., "Habitual Caffeine Use Is Associated with Heightened Cortisol Reactivity to Lab-Based Stress in Two Samples," Psychosomatic Medicine 86, no. 8 (2024): 730–37, https://doi.org/10.1097/PSY.0000000000001334.

[126] A. Caplin et al., "The Effects of Exercise Intensity on the Cortisol Response to a Subsequent Acute Psychosocial Stressor," Psychoneuroendocrinology 131 (September 2021): 105336, https://doi.org/10.1016/j.psyneuen.2021.105336.

[127] Alissa Shida et al., "Cortisol Levels After Cold Exposure Are Independent of Adrenocorticotropic Hormone Stimulation," PLOS One 15, no. 2 (2020): e0218910, https://doi.org/10.1371/journal.pone.0218910.

[128] Eva Fries et al., "The Cortisol Awakening Response (CAR): Facts and Future Directions," International Journal of Psychophysiology 72, no. 1 (2009): 67–73, https://doi.org/10.1016/j.ijpsycho.2008.03.014.

[129] Daniel J. Powell and Wolff Schlotz, "Daily Life Stress and the Cortisol Awakening Response: Testing the Anticipation Hypothesis," PLOS One 7, no. 12 (2012): e52067, https://doi.org/10.1371/journal.pone.0052067.

And just remember this: You want your cortisol and catecholamines relatively high in the morning and daytime, and you want them low in the evening and at night. Stack the tools described here to do just that.



STRESS CONTROL PROTOCOL

Consider Specific Supplements to Help Offset Excessive Stress

WHAT TO DO

- Find a combination of supplements that might help you combat severe or prolonged periods of stress. Remember: Behavioral protocols are the foundation of mental and physical health and performance, and that is especially true when it comes to stress control. It's also true that if your sleep duration and quality are not sufficient, no amount of pharmacology will help, at least not for long. Refer to chapter 1 for more on that topic. With that said, below is a list of supplements that have shown some efficacy for reducing various aspects of the stress response: Ashwagandha: 200–300 mg, 1 to 2 times per 24 hours, but taken only in the afternoon and/or evening.
- L-theanine: 100 mg or 200 mg, 1 to 3 times per day; if you drink caffeine regularly or are in a period of unusually high stress, take theanine with your caffeine.

- Phosphatidylserine: 400–800 mg, once per day anytime after 4:00 p.m.
- Rhodiola rosea: 200 mg, once or twice per day
- Proceed with caution. As always when discussing supplements or medication, it is important to consult with a physician. You need to be sure that you are not risking a dangerous interaction, that the product is reliable, and that a medical professional has information about what may be symptoms of an underlying health issue.

HOW IT WORKS

Certain supplements can help in times of excessive stress, such as when you are dealing with two or three nights of poor sleep due to an emotionally challenging circumstance, or if you've made behavioral changes but are still finding it hard to balance that autonomic seesaw. Alternatively, changes that occur with age can cause hormone surges or other shifts that can make you more vulnerable to stress.

On these occasions, certain supplements, taken alone or in combination, can help. The supplements in this protocol can increase parasympathetic nervous system activation and reduce sympathetic nervous system activation, either directly or indirectly, helping you to tilt that seesaw in the direction that is most beneficial to you. The studies described below are selected from the body of research on each supplement.

These summaries are not comprehensive but provide an overview of the evidence.

If you decide to explore supplements for limiting stress, I would suggest taking one of them and seeing how it works for you before adding another. That is going to be the most informative and cost-effective approach. (I am well aware that some people will opt to just dive in and take all four because they hate stress that much. Try to hold back.)

Ashwagandha

Ashwagandha, also known as Indian ginseng or winter cherry, is used as an adaptogen—a term that broadly refers to substances that improve a person's resistance to stress. A large review published in 2021 that included forty-one total studies in humans included seven that focused on stress and anxiety, enrolling a total of 491 adults who either reported being under high stress or were diagnosed with an anxiety disorder.[130] Ashwagandha dosages varied from 240 milligrams per day to 1,250 milligrams per day for an extract and up to 6,000 milligrams when administered as a powdered root. (Most ashwagandha capsules sold over the counter in the US contain 200–600 mg per capsule.) Taken

[130] Adrian L. Lopresti and Stephen J. Smith, "Ashwagandha (Withania somnifera) for the Treatment and Enhancement of Mental and Physical Conditions: A Systematic Review of Human Trials," Journal of Herbal Medicine 28, article no. 100434 (August 2021), https://doi.org/10.1016/j.hermed.2021.100434.

together, these studies showed that ashwagandha reduced stress and anxiety when compared to a placebo, and also reduced sleeplessness and fatigue. Cortisol levels in the blood were also lower among people taking this supplement, and it appeared that the cortisol-lowering effect of ashwagandha is likely the major basis of its effect on lowering stress and anxiety. A study of sixty adults found that those who took 225 milligrams or 400 milligrams per day of ashwagandha reported reductions in anxiety, depression, and food cravings compared to people randomized to a placebo.[131] One study found that men and women taking ashwagandha had a 22 percent and 25 percent reduction in cortisol, respectively, when they took ashwagandha for sixty days versus taking a placebo for sixty days.[132]

Research has identified several components that could contribute to the stress-reducing effects of ashwagandha. Most likely the effect is due to a "cocktail" of several different compounds in ashwagandha that work in parallel on various aspects of the hypothalamic-pituitary-adrenal axis, which, as you may remember from Sleep Protocol 1, is

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[131] A. Remenapp et al., "Efficacy of Withania somnifera Supplementation on Adults' Cognition and Mood," Journal of Ayurveda and Integrative Medicine 13, no. 2 (2022): 100510, https://doi.org/10.1016/j.jaim.2021.08.003.

[132] Adrian L. Lopresti et al., "An Investigation into the Stress-Relieving and Pharmacological Actions of an Ashwagandha (Withania somnifera) Extract: A Randomized, Double-Blind, Placebo-Controlled Study," Medicine 98, no. 37 (2019): e17186, https://doi.org/10.1097/MD.0000000000017186.

our main biological circuitry for mobilizing a physiological response to a stressor. The most potent of these mechanisms is the suppression of the signaling pathway for corticotropin (ACTH), a hormone that acts to stimulate the release of cortisol from your adrenals.[133] The ashwagandha-suppression of corticotropin (ACTH) happens through a multistep process that begins with compounds called withanolides (naturally occurring plant hormones). These compounds can lower corticotropin-releasing hormone (CRH) in the hypothalamus.[134] Limiting the amount of CRH in turn dampens the signaling that would normally trigger the release of cortisol. The fact that ashwagandha works "upstream" from the adrenal glands has a potential downside: Namely, it is likely to impact all the

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[133] Tobias Backström and Svante Winberg, "Central Corticotropin Releasing Factor and Social Stress," Frontiers in Neuroscience 7 (July 2013): 117, https://doi.org/10.3389/fnins.2013.00117; Michał Wiciński et al., "Can Ashwagandha Benefit the Endocrine System? A Review," International Journal of Molecular Sciences 24, no. 22 (November 2023); Günter Vollmer and Thomas Brendler, "Evaluation of Potential Hormonal Activities of Ashwagandha (Withania somnifera)," Phytotherapy Research, December 26, 2025, https://doi.org/10.1002/ptr.70155; Srikanta Pandit et al., "Effects of Withania somnifera Extract in Chronically Stressed Adults: A Randomized Controlled Trial," Nutrients 16, no. 9 (April 2024): 1293, https://doi.org/10.3390/nu16091293; Mateusz Haber et al., "Ashwagandha as an Adaptogen: Its Influence on Sleep Patterns, Stress Response, and Anxiety in Modern Life," Journal of Education, Health and Sport 68 (September 2024): 55327, https://doi.org/10.12775/JEHS.2024.68.55327.

[134] Hyeongyeong Kim et al., "Ashwagandha (Withania somnifera [L.] Dunal) Root Extract Containing Withanolide A Alleviates Depression-Like Behavior in Mice by Enhancing the Brain-Derived Neurotrophic Factor Pathway Under Unexpected Chronic Mild Stress," Journal of Ethnopharmacology 340 (2025): 119224, https://doi.org/10.1016/j.jep.2024.119224.

biology connected to CRH, which includes the gut and the immune system, among others. That's one of several reasons why I suggest not taking ashwagandha at dosages higher than 600 milligrams per day for more than a few weeks before taking at least a few weeks off. There is at least one report of someone taking a high dose (950 mg) of ashwagandha daily for a year and developing adrenal insufficiency.[135] People seem to vary quite a bit in their sensitivity to ashwagandha, so if you decide to try it in times of excess stress, I suggest starting with one dose per day, taken in the evening or an hour before bedtime at a dose of about 100 to 200 milligrams, and only increase the dosage, if needed, to find the minimal effective dose for you. Importantly, you don't want to suppress cortisol in the early part of your day. If you decide to take more, take ashwagandha in the afternoon around 3:00 to 4:00 p.m., and perhaps again an hour before bedtime in a dosage of 200 to 300 milligrams each.

As mentioned above, I suggest you do not take ashwagandha at doses exceeding 600 milligrams per day for periods of longer than a couple of weeks. While the reports are largely anecdotal, some people taking high doses of ashwagandha (700–1,200 mg) per day experience apathy, starting about

[135] Juveria Javid et al., "From Herbal Hope to Hormonal Havoc: Chronic Ashwagandha Use and HPA Axis Suppression," Endocrinology, Diabetes and Metabolism Case Reports 2026, no. 1 (January 2026).

three to four weeks after continuous intake. The reasons for this are not clear and may be similar to the case report mentioned above, but because efficacy has been shown at lower doses of 300 to 600 milligrams per day, I suggest people use it as a tool to augment stress reduction by taking about 300 milligrams on a regular basis as described above, or reserve its use to high-stress periods and take higher dosages (greater than 600 mg/day) only during those times. And don't let up on the behavioral practices for stress reduction.

A quick note about sourcing ashwagandha and other herbal adaptogens: The herbal supplement world is full of fake and contaminated products, perhaps even more so than for other types of supplements. Stick to major brands with third-party testing, especially ones that test for and remove metals from their products and ensure purity. Gaia Herbs has a good reputation for doing this and sells single-ingredient formulations. Barlow Herbals is another. I have no affiliation to either.

L-theanine

L-theanine is an amino acid present in green tea and has been recognized as one of its active compounds, alongside

caffeine, catechins, and other molecules.[136] Many studies of green tea have investigated a potential protective benefit against cancer and other chronic diseases,[137] with the evidence from laboratory and animal studies indicating that it may reduce the risk of obesity and type 2 diabetes, as well as heart disease[138] and breast cancer.[139]

Large-scale clinical trials exploring the efficacy of theanine in humans are still lacking. It's unlikely such studies will ever happen given their high cost and the fact that theanine is unlikely to be made into a pharmaceutical drug. That's just how it goes with most naturally occurring compounds.

The reason L-theanine is in this chapter is that it is widely used to reduce arousal. That's why nowadays L-theanine is sometimes added to energy drinks, and even to some coffee blends, to offset the jitters associated with drinking too much caffeine—presumably a clever trick on the part of the

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136 Muhammad Saeed et al., "Green Tea (Camellia sinensis) and L-Theanine: Medicinal Values and Beneficial Applications in Humans—a Comprehensive Review," Biomedicine & Pharmacotherapy 95 (November 2017): 1260–75, https://doi.org/10.1016/j.biopha.2017.09.024.

137 Raymond Cooper, "Green Tea and Theanine: Health Benefits," International Journal of Food Sciences and Nutrition 63, no. S1 (2012): 90–97, https://doi.org/10.3109/09637486.2011.629180.

138 Sabu M. Chacko et al., "Beneficial Effects of Green Tea: A Literature Review," Chinese Medicine 5 (April 6, 2010): 13, https://doi.org/10.1186/1749-8546-5-13.

139 Kathryn T. Kavanagh et al., "Green Tea Extracts Decrease Carcinogen-Induced Mammary Tumor Burden in Rats and Rate of Breast Cancer Cell Proliferation in Culture," Journal of Cellular Biochemistry 82, no. 3 (2001): 387–98, https://doi.org/10.1002/jcb.1164.

manufacturers that allows people to drink more of these beverages without feeling too strung out.

In one small, blinded, randomized study, thirty participants between the ages of thirty-six and sixty took L-theanine 200 milligrams per day or a placebo for four weeks.[140] The investigators, a team from Japan, used several standard measures to gauge depression, anxiety, and sleep quality. They reported that the group taking L-theanine had significantly fewer stress-related symptoms.

In another study, a group of researchers in Australia and the Netherlands enrolled thirty-four adults between the ages of eighteen and forty in a blinded study to measure stress response in reaction to a series of cognitive tests involving mental arithmetic, memory, and coordination.[141] Tasks like these can trigger stress because they challenge us. The tasks were presented following consumption of either a placebo or L-theanine as a nutrient drink. They found that the stress response, measured through the amount of salivary cortisol, was reduced compared to the placebo, an effect that appeared to last for several hours. In addition, the L-theanine

[140] Shinsuke Hidese et al., "Effects of L-Theanine Administration on Stress-Related Symptoms and Cognitive Functions in Healthy Adults: A Randomized Controlled Trial," Nutrients 11, no. 10 (2019): 2362, https://doi.org/10.3390/nu11102362.

[141] David J. White et al., "Anti-Stress, Behavioural and Magnetoencephalography Effects of an L-Theanine-Based Nutrient Drink: A Randomised, Double-Blind, Placebo-Controlled, Crossover Trial," Nutrients 8, no. 1 (2016): 53, https://doi.org/10.3390/nu8010053.

group showed an increase in alpha brain waves, suggesting a more relaxed state of mind. That being said, cognitive performance was no different between the L-theanine and placebo groups. The reduced stress reaction may be beneficial for other reasons, but it's useful to know that theanine didn't make people better at the tasks they were given.

In terms of the underlying mechanisms, L-theanine research has found that this compound indirectly increases GABA—the major inhibitory neurotransmitter in the brain—and dopamine.[142]

Theanine taken in the hour before bedtime is part of the sleep stack I describe in chapter 1. There, I warned that people who have especially vivid dreams not take theanine. I suggest that people who want to try theanine to reduce their stress do so by taking about 200 milligrams of it in the early afternoon and perhaps another 100 to 200 milligrams again in the evening, especially if you had caffeine in the early afternoon.

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[142] Ryohei Kimura and Toshiro Murata, "Influence of Alkylamides of Glutamic Acid and Related Compounds on the Central Nervous System. I. Central Depressant Effect of Theanine," Chemical and Pharmaceutical Bulletin 19, no. 6 (1971): 1257–61, https://doi.org/10.1248/cpb.19.1257.

Phosphatidylserine

Phosphatidylserine is a type of phospholipid. It is naturally found in the membranes of neurons, and comprises about 10 to 20 percent of the total phospholipid content of the brain—an enormous percentage when one considers that much of the nonliquid content of the brain is phospholipids. Naturally occurring phosphatidylserine plays a role in cognitive function,[143] and the loss of phosphatidylserine in neurons known to occur in later adulthood may partially explain age-related (non-dementia) cognitive decline.

What's more, in its natural state in our brains, this compound is known to decrease the stress response.[144] As a supplement, phosphatidylserine has shown some benefit in older adults who experience chronic stress. There is some evidence phosphatidylserine taken late in the day may preferentially reduce evening and nighttime cortisol,[145] which may explain how it indirectly supports sleep.

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[143] Hee-Yong Kim et al., "Phosphatidylserine in the Brain: Metabolism and Function," Progress in Lipid Research 56 (October 2014): 1–18, https://doi.org/10.1016/j.plipres.2014.06.002.

[144] Palmiero Monteleone et al., "Effects of Phosphatidylserine on the Neuroendocrine Response to Physical Stress in Humans," Neuroendocrinology 52, no. 3 (1990): 243–48, https://doi.org/10.1159/000125593.

[145] Katja Schmidt et al., "A Lecithin Phosphatidylserine and Phosphatidic Acid Complex (PAS) Reduces Symptoms of the Premenstrual Syndrome (PMS): Results of a Randomized, Placebo-Controlled, Double-Blind Clinical Trial," Clinical Nutrition ESPEN 24 (April 2018): 22–30, https://doi.org/10.1016/j.clnesp.2018.01.067.

In one investigation, from 2004, participants took a phosphatidylserine supplement at a dose of 400 milligrams, 600 milligrams, or 800 milligrams, or a placebo (twenty people per group) and then completed a type of social stress test that involves performing in front of an audience.[146] Data collected on biological indicators of the stress response revealed a significant decrease in these symptoms among the 400-milligram group, but not among those taking the higher doses (or the placebo). It's hard to interpret the effect of higher doses, but this is not uncommon in pharmacology studies. Many studies of stress and cognitive performance show an inverted U-shaped response, where low- and high-dose conditions yield similar or no effects, but mid-range dosages do. Such are the complexities of biology. Another study found no link between phosphatidylserine and measures of a stress response across all sixty participants, but the researchers did observe an effect among participants with high chronic stress levels.[147] Initially, this group had a diminished cortisol rise in response to a social stress test. That may sound like a good thing but it's not. A blunted cortisol response is often a sign of underlying issues, such as

[146] J. Hellhammer et al., "Effects of Soy Lecithin Phosphatidic Acid and Phosphatidylserine Complex (PAS) on the Endocrine and Psychological Responses to Mental Stress," Stress 7, no. 2 (2004): 119–26, https://doi.org/10.1080/10253890410001728379.

[147] Juliane Hellhammer et al., "Omega-3 Fatty Acids Administered in Phosphatidylserine Improved Certain Aspects of High Chronic Stress in Men," Nutrition Research 32, no. 4 (2012): 241–50, https://doi.org/10.1016/j.nutres.2012.03.003.

depression.[148] Treatment with phosphatidylserine appeared to restore the healthy rise in cortisol in response to the test, meaning it seemed to recalibrate the stress response rather than stop it. I say all this not to confuse you but to remind you that though the best model for your autonomic nervous system might be a seesaw, we need to consider the "hinge" of that seesaw (metaphorically speaking). How it bends, or sticks, matters. And compounds like phosphatidylserine may change people's stress response by adjusting mechanisms analogous to how you might treat a hinge, rather than merely increasing or decreasing cortisol or adrenaline. Finally, in a six-week study from 2014 of seventy-five men randomized to phosphatidylserine (at a dose of 200 or 400 mg per day) or a placebo, the higher dose of the supplement appeared to lower the stress response among participants with chronic high stress.[149]

If you opt to try phosphatidylserine for stress control, I recommend starting with a low dose of 100 to 200 milligrams, taken a few hours before bedtime, and increasing

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[148] Elizabeth J. Susman, "Psychobiology of Persistent Antisocial Behavior: Stress, Early Vulnerabilities and the Attenuation Hypothesis," Neuroscience & Biobehavioral Reviews 30, no. 3 (2006): 376–89, https://doi.org/10.1016/j.neubiorev.2005.08.002.

[149] Juliane Hellhammer et al., "A Soy-Based Phosphatidylserine/Phosphatidic Acid Complex (PAS) Normalizes the Stress Reactivity of Hypothalamus-Pituitary-Adrenal-Axis in Chronically Stressed Male Subjects: A Randomized, Placebo-Controlled Study," Lipids in Health and Disease 13 (July 2014): 121, https://doi.org/10.1186/1476-511X-13-121.

the dose only as needed. I know many people who like its effects on lowering stress late in the day, and for sleep.

Rhodiola rosea

Rhodiola rosea is a yellow-flowered plant that grows in northern regions of Europe and Asia and on the Atlantic coast of North America. It's long been part of traditional medicine practices in Europe and Asia, used to address fatigue, mild anxiety, and symptoms of burnout.[150] It is considered an adaptogen, because it alters the subjective experience of effort, apparently by adjusting the threshold for perceived effort, and by increasing effort endurance in physical and mental exertion. Subjectively, people taking Rhodiola rosea report less post-physical-exertion-related fatigue and that mental stressors feel less stressful, meaning they can lean into sustained effort without feeling exhausted. Indeed, Rhodiola rosea has been shown to reduce the symptoms of burnout, which is a multidimensional feature of chronic stress that includes the "wired but tired" feeling that is a common consequence of prolonged periods (days, weeks, etc.) of stress and altered sleep-wake cycles. European medical authorities approved the use of Rhodiola

[150] Emilija Ivanova Stojcheva and José Carlos Quintela, "The Effectiveness of Rhodiola rosea L. Preparations in Alleviating Various Aspects of Life-Stress Symptoms and Stress-Induced Conditions—Encouraging Clinical Evidence," Molecules 27, no. 12 (2022): 3902, https://doi.org/10.3390/molecules27123902.

rosea for short-term relief of stress in 2012[151] and data on this connection have been slowly accruing.

One study examined the effects of Rhodiola rosea among fifty-six physicians working night shifts for fourteen days.[152] Signs of mental fatigue were tested before and after the night shift for three two-week stretches. The investigators reported an improvement in tests examining memory, concentration, perception, and other aspects of brain function during the first two-week administration of Rhodiola rosea.

Another trial, from Russia, randomized 161 military cadets to one dose of Rhodiola rosea per day at either 370 milligrams or 555 milligrams, or placebo, and found a reduction in fatigue among both supplement groups.[153] And in another investigation, a phase 3 clinical trial at Uppsala University, in Sweden, sixty people were randomized to receive four daily doses of Rhodiola rosea, for a total of 576

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[151] "Assessment Report on Rhodiola rosea L., Rhizoma et Radix," Committee on Herbal Medicinal Products, European Medicines Agency, March 27, 2012, https://www.pharmacompass.com/pAssets/pdf/pubchem/rhodiola-rosea-l.pdf.

[152] V. Darbinyan et al., "Rhodiola rosea in Stress-Induced Fatigue—a Double Blind Cross-Over Study of a Standardized Extract SHR-5 with a Repeated Low-Dose Regimen on the Mental Performance of Healthy Physicians During Night Duty," Phytomedicine 7, no. 5 (2000): 365–71, https://doi.org/10.1016/S0944-7113(00)80055-0.

[153] V. A. Shevtsov et al., "A Randomized Trial of Two Different Doses of a SHR-5 Rhodiola rosea Extract Versus Placebo and Control of Capacity for Mental Work," Phytomedicine 10, no. 2–3 (2003): 95–105, https://doi.org/10.1078/094471103321659780.

milligrams per day, or four placebo tablets.[154] The participants completed several questionnaires gauging quality of life, depression, fatigue, and attention. Both groups experienced improvements in overall mental health and reduced feelings of burnout. But in the group taking Rhodiola rosea, the reduction in fatigue was more pronounced, and linked to improvements in performance and concentration.

Rhodiola rosea is among the better-studied supplements for stress control, and we have copious evidence pointing to its usefulness in this context. With regard to the underlying mechanism, Rhodiola rosea contains numerous phytochemicals (compounds made by plants) that could contribute to its benefits, including by interacting with cortisol pathways[155] and by activating heat-shock proteins,[156] which our cells make in response to stressful conditions such as the excessive heat or cold, the presence of heavy metals, or an infection. Researchers are still

[154] Erik M. G. Olsson et al., "A Randomised, Double-Blind, Placebo-Controlled, Parallel-Group Study of the Standardised Extract SHR-5 of the Roots of Rhodiola rosea in the Treatment of Subjects with Stress-Related Fatigue," Planta Medica 75, no. 2 (2009): 105–12, https://doi.org/10.1055/s-0028-1088346.

[155] A. Panossian et al., "Rosenroot (Rhodiola rosea): Traditional Use, Chemical Composition, Pharmacology and Clinical Efficacy," Phytomedicine 17, no. 7 (2010): 481–93, https://doi.org/10.1016/j.phymed.2010.02.002.

[156] Aaron Hernández-Santana et al., "A Rhodiola rosea Root Extract Protects Skeletal Muscle Cells Against Chemically Induced Oxidative Stress by Modulating Heat Shock Protein 70 (HSP70) Expression," Phytotherapy Research 28, no. 4 (2014): 623–28, https://doi.org/10.1002/ptr.5046.

investigating the exact mechanisms tied to the stress-reducing effects, as well as its apparent contribution to physical strength and endurance.[157] The picture emerging is that Rhodiola rosea buffers against stress by calibrating the amount of adrenaline that is released in moments of high physical exertion and high emotional stress.

I recommend one of two approaches to trying Rhodiola rosea. The first is to take it thirty to sixty minutes before a known stressor, like a hard workout or big exam or study session or public speaking engagement—basically any event that you know will require an intense and prolonged effort. I take 120 to 240 milligrams of Rhodiola rosea before hard resistance training workouts, which is per the standard protocol used in most peer-reviewed studies of Rhodiola rosea for physical exertion. The second way to take Rhodiola rosea is during periods of chronic stress, when you can't get as much sleep as you would like and/or when your output is greater than usual. Think final exam week, the days of long work hours before a vacation, the vacation with your in-laws where you are coordinating everything, et cetera. In those times, I suggest taking 100 to 120 milligrams (the typical capsule dose of Rhodiola rosea in most supplements) three times daily (morning, afternoon, and evening). It can be

[157] Tyler D. Williams et al., "Effects of Short-Term Golden Root Extract (Rhodiola rosea) Supplementation on Resistance Exercise Performance," International Journal of Environmental Research and Public Health 18, no. 13 (2021): 6953, https://doi.org/10.3390/ijerph18136953.

taken with or without food. I would not suggest taking Rhodiola rosea multiple times throughout the day for more than three to four weeks without taking a break of at least two weeks, as all the biological pathways Rhodiola rosea is known to impact are subject to feedback loops and downregulation. Use 100-milligrams-plus daily dosages of Rhodiola rosea as a short-term tool to combat stress and burnout, not as a mainstay of your supplement regimen. And I know I sound like a broken record, but don’t just pile on supplements for managing stress. You need—and now have—multiple behavioral tools to reduce stress. Use them!